The Intramonth Momentum Cycle*

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Abstract

U.S. equity momentum returns concentrate in a six-trading-day window ending four trading days before month-end. This pre-turn-of-the-month (PreTOM) window earns 63% of the momentum premium over 1980–2025 and all of it since 2002 (our sample mid-point). The concentration is driven by loser stocks, which face disproportionate selling pressure during the PreTOM window, as investors sell stocks to meet month-end cash obligations. Trade-level data show that seller-initiated imbalance rises among losers during PreTOM, while mutual funds facing outflows disproportionately liquidate loser positions. Settlement reforms provide corroborating evidence: the U.S. shift from T+2 to T+1 and the European shift from T+3 to T+2 each move loser stock price pressure one trading day closer to month-end. The PreTOM pattern appears in 19 developed markets. These findings identify month-end liquidity management as a substantial source of momentum returns, distinct from leading behavioral and risk-based explanations.

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1 Introduction

Momentum, the tendency of past winners to outperform past losers, is among the most pervasive asset-pricing anomalies. Since Jegadeesh and Titman (1993), it has been documented across equity markets and has become a cornerstone of systematic investing.[1] Yet its underlying mechanisms remain contested. Behavioral explanations emphasize gradual information diffusion and investor underreaction, while risk-based theories attribute momentum to compensation for exposure to priced risks.[2] We identify a large, systematic component of momentum returns that arises from institutional liquidity demand around month-end. This component is concentrated in loser stocks and is largely distinct from existing explanations of momentum.

In the U.S., momentum returns concentrate in the six trading days ending four trading days before month-end, which we call the pre-turn-of-the-month window (PreTOM). Over 1980–2025, a dollar invested in a value-weighted winners-minus-losers (WML) strategy only during PreTOM grows to \$18.78, compared with \$2.37 when invested during the remainder of the month (Figure 1). The higher return during PreTOM comes almost entirely from the short leg: loser stocks underperform the market by 8 basis points per day during PreTOM. Momentum returns have also become increasingly concentrated in PreTOM over time. Before 2002, momentum earns positive returns both inside and outside the window; since 2002, the entire premium has accrued during PreTOM.[3,4] The PreTOM concentration is not specific to U.S. equities. Across 19 developed markets, losers underperform their local market by 3.2 basis points per day more during PreTOM than on other days. Underperformance is present in 14 of the 19 markets.

Our research builds on Etula et al. (2020), who show that institutions' month-end liquidity needs generate predictable market-level selling pressure that ends four trading days before month-end. These needs arise from cash obligations associated with pension payments, div-

[1] See Gupta and Doole (2025).

[2] See, e.g., Hong and Stein (1999) and Daniel et al. (1998) for behavioral explanations; see Johnson (2002), Chordia and Shivakumar (2002), and Cochrane et al. (2008) for risk-based explanations.

[3] Throughout we use the standard portfolio construction of the momentum strategy, sorting on returns over months -12 to -2, skipping a month, then holding for a month. The momentum return concentration is equally stark at a six-month holding horizon. The overlapping Jegadeesh and Titman (1993) six-month holding horizon momentum strategy earns 72.6 basis points per month, with 55.0 on these six days and 17.5 on the others.

[4] Bhattacharya et al. (2017) document the post-2000 attenuation of U.S. momentum.

idends, interest, and salaries. We show that this institutional selling pressure to obtain month-end liquidity falls disproportionately on momentum losers, generating unusually low loser returns and, consequently, high momentum returns during PreTOM.[5]

Several findings support this liquidity-based interpretation. Transaction-level data from the Trade and Quote (TAQ) database show that seller-initiated order imbalance is disproportionately high for loser stocks and increases during PreTOM. Settlement reforms provide a test of the mechanism's timing. Shorter settlement periods allow investors to sell later while still receiving cash before payment deadlines. Following the U.S. transition from T+2 to T+1 settlement in May 2024, loser returns improve on the fourth trading day before month-end and deteriorate on the third. The European transition from T+3 to T+2 settlement in October 2014 also moved loser price pressures one trading day closer to month-end, from the fifth trading day before month-end to the fourth. Price pressures moved in the predicted direction in 10 of the 13 affected markets. These changes in the timing of price pressure are consistent with the interpretation that month-end liquidity needs drive loser selling during PreTOM and that settlement conventions determine when investors must sell to meet their payment deadlines. Daily mutual fund flows support the liquidity-based interpretation: outflow pressure decreases monotonically with a stock's momentum decile rank.

Mutual fund holdings data provide evidence of a complementary, selection-based channel that creates selling pressure on losers. Within the same fund, loser positions are more likely to be closed out between consecutive quarterly holdings reports when net outflows are higher. This relation holds in every position-weight quintile and survives controls for position weight, stock liquidity, and stock size. More generally, funds facing outflows shrink loser positions faster than their other holdings. Together, the trading, flow pressure, and holdings evidence shows that liquidity-driven selling falls disproportionately on losers.

The price pressure does not unwind at the turn of the month. A one-standard-deviation increase in incremental PreTOM selling is associated with a 131.6-basis-point loss during the PreTOM window and only an 8.7-basis-point rebound over the following seven trading

[5] Etula et al. (2020) study a fixed five-trading-day pre-turn-of-the-month window ending four days before month-end. We use a six-day window, which begins one trading day earlier, as it captures the deepest part of the market index decline in our sample. Using other nearby windows gives similar results. Because the PreTOM window's end is tied to the settlement cycle, we also tried shifting it to match the settlement rule in force in each period (T+5 before June 1995, T+3 until September 2017, T+2 until May 2024, T+1 since). The estimates are essentially unchanged.

days. Thus, more than 90 percent of the loss remains a week later. This initial rebound of the most heavily sold losers also disappears within days. The loser portfolio as a whole does bounce back about 30 percent of its 47.9-basis-point loss at the turn of the month, but this rebound is also temporary and is gone by the ninth trading day of the following month. Stocks that newly enter the loser decile stay below their pre-PreTOM price level for about 9 months on average.

Because momentum could earn positive returns on any six days, we also measure the PreTOM returns against returns on ordinary trading days. Against them PreTOM delivers an abnormal momentum return of 45.3–46.6 basis points per month, equivalent to roughly half the average momentum premium. The benchmark days themselves are subject to some of the same liquidity-driven selling pressure, making this estimate a conservative measure of PreTOM's contribution.

Historical evidence shows that this PreTOM concentration in returns was largely absent in the first half of the twentieth century, but strengthened alongside the increase in institutional ownership (Blume and Keim, 2012). This historical pattern further points to institutional cash management as an important source of modern momentum. Loser underperformance and elevated momentum returns also emerge during other episodes of systemic liquidity demand, in 2008 and March 2020. These episodes suggest that liquidity-driven selling affects momentum returns more broadly, beyond the regular month-end liquidity cycle.

Investors who need cash sell positions that are most attractive to give up. We call such stocks dispensable. Following the standard momentum literature, we define losers by the conventional 12-2 sort. The disproportionate selling of losers during month-end liquidity events indicates that the loser portfolio contains stocks that investors regard as dispensable. But loser status alone does not define dispensability. Within the loser portfolio, we find that a stock's distance from its 52-week high provides a good proxy for dispensability. Several features make those stocks particularly attractive to sell when cash is needed: stocks far below their highs exhibit salient underperformance (George and Hwang, 2004), are likely to carry short-term unrealized losses that are most valuable for tax purposes, and tend to pay less dividend income. Seller-initiated imbalance among losers rises monotonically with distance from the 52-week high, and the losers farthest below their highs underperform those nearest them during PreTOM, but not on other days. This proxy for dispensability therefore helps explain which of the loser stocks bear the liquidity-driven selling.

Our findings are distinct from most of the leading theories of momentum. Disposition-based explanations (Grinblatt and Han, 2005; Frazzini, 2006) attribute return continuation to investors' reluctance to realize losses, which delays the incorporation of negative information. Our evidence instead shows that losers face unusually high selling pressure when investors need cash, the opposite of a reluctance to sell. These findings do not rule out a role for the disposition effect in momentum generally, but it identifies a distinct mechanism behind the PreTOM component of momentum.

Other theories of momentum emphasize slow information diffusion, underreaction, or overreaction to information (Hong and Stein, 1999; Daniel et al., 1998; Andrei and Cujean, 2017). Our mechanism instead links the timing and cross-sectional incidence of momentum returns to cash obligations, settlement conventions, and the dispensability of loser positions. It complements the work connecting capital flows to momentum, such as Lou (2012) and Vayanos and Woolley (2013), and research on slow-moving capital and infrequent investor trading (Mitchell et al., 2007; Duffie, 2010; Bogousslavsky, 2016). The distinction is that we identify a recurring month-end liquidity channel that predicts both when selling of losers occurs and which loser stocks bear it.[6,7] Unlike risk-sharing models in which both legs are affected (Cochrane et al., 2008), this channel operates on losers alone.

Our paper also contributes to the literature on calendar effects and institutional trading. Ariel (1987), Lakonishok and Smidt (1988), and Ogden (1990) establish high returns at the beginning of the month. Etula et al. (2020) document high returns in the last four days of the month, and poor market returns before the turn of the month. Related research examines more generally recurring seasonalities in stock returns (Heston and Sadka, 2008, 2010), intraday and overnight returns (Lou et al., 2019; Bogousslavsky, 2021; Barardehi et al., 2026), same-weekday continuation (Keloharju et al., 2016, 2021; Da and Zhang, 2025), and predictable price pressure associated with dividend cash flows (Hartzmark and Solomon, 2013, 2025). Our evidence extends this research by identifying a distinct monthly pattern in momentum, concentrated in the loser leg. The weekly momentum clock of Da and Zhang

[6] Grinblatt et al. (1995) and Sias (2004) also document momentum-chasing behavior by institutions. Pitkäjärvi et al. (2020) in turn link time-series momentum to equity fund flows and bond market returns.

[7] Demand-system frameworks (Koijen and Yogo, 2019; Gabaix and Koijen, 2021) provide the infrastructure for studying how investor flows shape asset prices, and our results identify a specific flow-driven mechanism within that framework. Lou and Polk (2022) examine how institutional arbitrage in momentum strategies itself affects returns.

(2025) does not explain our monthly pattern. It also differs from the turn-of-the-month rebalancing between stocks and bonds studied by Harvey et al. (2025) and the quarter-end concentration in institutional momentum trading documented by Sias (2007).

Finally, our results relate to the asymmetry of anomaly returns and momentum crashes. Prior work shows that anomaly profits, including momentum, often concentrate in the short leg (Stambaugh et al., 2012; Hong et al., 2000; Avramov et al., 2013). We show that this short-leg concentration also has a pronounced calendar dimension. We show that the PreTOM return premium is not explained by the timing of momentum crashes: crash days occur within the PreTOM window at approximately their unconditional calendar frequency. Instead, we show that crashes cluster at month-start. The PreTOM premium also survives excluding major crash episodes and progressively trimming the worst daily returns. The concentration therefore reflects a broad feature of the return distribution rather than a few extreme observations.

The remainder of the paper describes the data and PreTOM window, shows that loser losses in the window persist and quantifies the window's contribution to momentum, tests the settlement-calendar predictions, documents selling pressure and its cross-sectional determinants, and examines the implications for the disposition-based and other explanations of momentum.

2 Data and the PreTOM Window Returns

2.1 Data

We construct daily momentum decile portfolios from CRSP using fixed monthly sorts. All NYSE, AMEX, and NASDAQ common stocks are ranked into deciles on cumulative returns over months -12 through -2 using NYSE breakpoints, and assignments are held constant within each calendar month. The strategy is that of Jegadeesh and Titman (1993); the decile construction with NYSE breakpoints follows Carhart (1997) and the conventions of Kenneth French's data library. WML is the value-weighted (VW) return of the top decile (D10) minus the bottom decile (D1), with lagged capitalization weights. We also report equal-weighted (EW) variants. Daily Fama–French factors and the risk-free rate are from Kenneth French's data library. Our baseline sample begins in 1980, following the institutional-flow literature

(Lewellen, 2011; Lou, 2012; Coval and Stafford, 2007).

Other datasets each speak to a different part of the mechanism: order flow to the direction of trading, mutual-fund holdings and flows to which positions funds sell, and international returns to whether the pattern is specific to the U.S. The stock-level panel contains 53.3 million CRSP stock-days over 1980–2025, merged with decile assignments and bid-ask quotes. We construct the same portfolio sorts back to 1927 for the historical analysis. Delisting returns are from the CRSP event file.

Trade direction comes from the WRDS Intraday Indicators (Lee–Ready classification of the NYSE Trade and Quote database, TAQ), 2003–2022, from which we compute daily net selling pressure, sell volume minus buy volume over total volume. Opening prices for the intraday-overnight decomposition are from CRSP, 1992–2024. Mutual-fund holdings are from Thomson S12, matched through MFLINKS to CRSP monthly net assets and returns. We construct implied net flows and aggregate them over each holdings-reporting interval for 1991–2024. Daily fund returns, computed from net asset values and distributions in the same database, begin in September 1998 and are used in Internet Appendix Section IA.24.

Daily fund flows are from Morningstar, 2008–2023, for U.S. equity open-end funds, excluding exchange-traded funds. Morningstar reports each fund's total flow on every one of its share classes, so we keep one observation per fund and day; summing across classes would count the same flow several times. We link funds to Thomson S12 holdings through CUSIPs, the CRSP mutual fund file and MFLINKS. When estimating stock-specific flow pressure, we multiply each fund's daily flow, as a fraction of its assets, by the fraction of the stock's shares outstanding that the fund holds at the most recent quarter-end that is at least three months before the flow date, and sum across funds. This equals dividing the fund's flow-scaled dollar position by the stock's market capitalization, with both measured at the report date (Wardlaw, 2020). Morningstar's daily flow coverage is highly incomplete before 2008.

For international evidence we construct analogous decile portfolios for 20 developed markets from Compustat Global, 1991–February 2026, using size-conditional breakpoints within country and regional pooling for small markets (Internet Appendix Section IA.6). The 19 markets outside Japan form the principal pooled sample. Japan is reported separately, following the literature on the absence of a momentum premium there. The same panel supplies the thirteen European markets that form the T+3-to-T+2 treatment sample for the October

2014 reform.[8]

2.2 Calendar Windows and Notation

Let τ be the upcoming month-end and τ' the previous month-end. We define the PreTOM window as $[\tau-9, \tau-4]$ and the Rest window as the union of $[\tau'+1, \tau-10]$ and $[\tau-3, \tau]$, the two within-month periods of main interest. For control we also define Mid, $[\tau'+4, \tau-10]$, and Early, $[\tau'+4, \tau'+9]$.

We use lowercase r to denote a daily return on a stock, a portfolio, or a factor, with r_t^{mkt} the daily value-weighted market return from Kenneth French's data library and r_t^f the risk-free rate. Uppercase R denotes a return cumulated over the interval named in its superscript. Subscripts identify stock i, day t, and month m. $D_{i,t} \in \{1, \ldots, 10\}$ is stock i's momentum decile in the month containing day t. $\text{BAS}_{i,m-1}$ is the stock's average daily percentage bid-ask spread over the prior calendar month.

U.S. equity settlement shortened from T+5 to T+3 in June 1995, from T+3 to T+2 on September 5, 2017, and from T+2 to T+1 on May 28, 2024 (Securities and Exchange Commission, 2017, 2023). Thirteen European markets in our sample shortened settlement from T+3 to T+2 in October 2014. Section 4 uses the 2024 U.S. and 2014 European transitions as experiments.[9]

2.3 The PreTOM Window Returns

Figure 1 shows how concentrated momentum returns are in the PreTOM window. Over 1980–2025, one dollar invested in WML only during the six trading days $[\tau-9, \tau-4]$ grows to \$18.78, against \$2.37 when invested only during the rest of the month. Figure 2 shows that the concentration comes mainly from the loser leg: losers underperform sharply during PreTOM, while winner returns change much less.

The loser leg develops a pre-month-end trough, much deeper than the market's. The market displays the turn-of-the-month pattern similar to that identified in Etula et al. (2020), with negative returns from $\tau-9$ to $\tau-4$ and positive returns at month-end, $\tau-3$ to τ, and

[8] Construction details for every dataset are in Internet Appendix Sections IA.1 and IA.6.

[9] We also considered using the U.S. 2017 settlement change as a third experiment. The estimate has the predicted sign but is small and insignificant, so we focus on the two other experiments (Section 4.1 discusses why the 2017 change may have had less effect).

the first days of the next month, $\tau+1$ to $\tau+3$, but losers follow a much steeper profile. Their return begins falling relative to the market around the eighth trading day of the month, $\tau'+8$, and stays weak through $\tau-1$, averaging -6.4 basis points per day over $[\tau'+8, \tau-1]$ and -8.0 over $[\tau-9, \tau-4]$. The six-day PreTOM window covers the core of that decline. Nearby windows give similar estimates (Internet Appendix Table IA.20), and Internet Appendix Figure IA.10 plots the full daily profile.

The trailing edge of the PreTOM window, $\tau-4$, sits two trading days before the last sale that would settle by month-end under T+2, and we find that it moves when the settlement cycle shortens (Section 4). The leading edge, $\tau-9$, is where the loser returns turn down more steeply. Individual-day estimates are noisy, however, with standard errors of 4 to 6 basis points, so we use the six-day PreTOM window in most of our analysis, as it is more informative than any single day.

Our results do not depend on the exact boundaries of the PreTOM window. Internet Appendix Table IA.20 reports returns for windows $[\tau-k, \tau-4]$ for $k = 4$ to 12 (Panel A), and for every six-day block in the second half of the month (Panel B). Estimates form a plateau around $[\tau-9, \tau-4]$. Panel C of that table asks whether a window chosen by searching the data outperforms ours out of sample. It selects windows based on the 1980–2000 sample and evaluates them on 2001–2025. The best $[\tau-k, \tau-4]$ window in the first period is $[\tau-8, \tau-4]$, the window of Etula et al. (2020), and it earns 10.45 basis points per day in 2001–2025, against 9.06 for our window over the same years. The best unrestricted six-day block, $[\tau-11, \tau-6]$, earns 6.07. The unrestricted search loses much of its return out of sample, while windows ending at $\tau-4$ hold up.

2.4 Portfolio and Stock-Level Estimates

Table 1 compares daily portfolio returns during PreTOM and on other days. Individual-leg returns are measured relative to the market. Losers earn -0.87 basis points per day on other days and -7.98 in the PreTOM window, a differential of -7.12 ($t = -2.79$, Newey–West with 21 lags). Winners earn 1.52 and 2.17, a differential of 0.65 ($t = 0.39$). WML earns 2.39 and 10.15, a differential of 7.76 ($t = 2.28$). Figure 2 shows the same comparison using returns compounded within each window and averaged across months; the estimates differ

by less than one basis point.[10]

The loser leg therefore contributes an abnormal return of 7.1 basis points per day during PreTOM, or about 42.7 basis points over the six-day window.

At the stock level we estimate

$$r_{i,t} - r_t^{mkt} = \beta_1 \,\text{Loser}_{i,t} + \beta_2 \,\text{Loser}_{i,t} \times \text{PreTOM}_t + \mu_i + \delta_t + \varepsilon_{i,t}, \tag{1}$$

where $\text{Loser}_{i,t}$ equals one for bottom-decile stocks and PreTOM_t equals one for days in the PreTOM window, and μ_i and δ_t are stock and day fixed effects. Standard errors are two-way clustered by stock and day.

Results show that loser stocks underperform disproportionately during the PreTOM window (Table 2). With stock and day fixed effects (columns 2 and 5), the value-weighted estimate of β_2 is -7.15 basis points per day ($t = -3.08$) and the equal-weighted estimate is -2.55 ($t = -1.85$). Both are within a tenth of a basis point of the estimates with day fixed effects alone, -7.08 and -2.51, so the PreTOM window effect does not come from permanent differences across stocks. The negative effect for value-weighted losers at PreTOM is present in both halves of the sample (-5.15, $t = -2.07$, in 1980–2001 and -9.02, $t = -2.35$, in 2002–2025; Internet Appendix Table IA.23), and in the 19 developed market panel regression in Section 2.5. Controlling for each stock's 60-month rolling CAPM market beta changes the value-weighted coefficient only from -7.30 to -7.02 basis points per day (Internet Appendix Table IA.17). The PreTOM loser underperformance is stronger in value-weighted portfolios. This is consistent with our mechanism, since institutions hold mostly large stocks, and with Etula et al. (2020), who find that pre-month-end price pressure is strongest in large, liquid stocks.

Columns 3 and 6 of Table 2 condition on the stock's liquidity before the PreTOM window opened, adding the logarithm of the stock's mean quoted spread over the preceding calendar month together with its interactions. The PreTOM effect does not weaken. The value-weighted estimate is -8.04 basis points per day ($t = -2.82$) and the equal-weighted estimate is -4.35 ($t = -2.83$), larger than the -2.55 of column 5, which is estimated on the full sample without the spread controls. Holding the sample fixed, the equal-weighted estimate is -2.88

[10] The compounded loser PreTOM window daily mean is -7.92 ($t = -3.72$); the daily mean is -7.98 ($t = -3.75$).

without the spread controls and −4.35 with them. The PreTOM loss is concentrated in stocks that are cheap to trade: on PreTOM days, returns are higher for loser stocks with wider spreads (PreTOM × Spread = 1.54, $t = 2.35$).[11]

2.5 International Evidence

We test whether the PreTOM loser effect extends beyond U.S. equities using daily momentum decile portfolios for 20 developed markets from Compustat Global, 1991–February 2026 (Hong Kong and Switzerland from 1993). Japan is reported separately, following the literature that documents the absence of a momentum premium there. No screen on realized momentum is applied to the remaining 19 markets.[12] Within each country-month, we rank stocks into momentum deciles on cumulative returns over months −12 through −2, value-weight by lagged market cap, and subtract the country's own value-weighted market return, computed from that country's stocks in the same panel.

We pool country-day observations and regress the loser-decile market-adjusted return on a PreTOM indicator with country fixed effects:

$$r_{c,t} - r_{c,t}^{mkt} = \alpha_c + \beta \, \text{PreTOM}_t + \varepsilon_{c,t}, \tag{2}$$

where $r_{c,t} - r_{c,t}^{mkt}$ is the day-t value-weighted loser-decile return in country c in excess of that country's own market return $r_{c,t}^{mkt}$, PreTOM_t is the indicator for trading days $[\tau-9, \tau-4]$ (Rest days are the omitted reference), and α_c are country fixed effects. Standard errors are clustered by calendar month to allow for correlation across countries within a month. In every country, PreTOM comprises local trading days $[\tau_{c,m}-9, \tau_{c,m}-4]$, where $\tau_{c,m}$ is the last local trading day of month m. We apply this same rule across countries rather than tailoring it to each market's data or settlement regime. This avoids look-ahead bias and is conservative, since in markets with longer settlement cycles, such as the European T+3 markets before October 2014, the fixed window could misplace part of the selling pressure.

[11] Untabulated; same specification as Table 2, column 6.

[12] Countries, sample periods, and methodology details (size-conditional breakpoints, regional pooling for small markets) are described in Internet Appendix IA.6. Canada is omitted because Compustat Global coverage is limited; Japan's lack of a momentum premium is documented in Fama and French (2012), who find average winner-minus-loser returns close to zero there, and Asness et al. (2013), who find that the premium is positive but insignificant.

The PreTOM concentration extends beyond U.S. equities. Across the 19 developed markets (Table 3), WML earns 10.0 basis points per day during PreTOM and 4.5 on other days, a differential of 5.5 basis points ($t = 2.36$). The loser leg contributes materially. Losers underperform their own country's market by an additional 3.2 basis points per day during PreTOM ($t = -2.24$), and the differential is negative in 14 of the 19 markets.[13] The corresponding winner differential is smaller, +2.1 basis points, and is not statistically distinguishable from zero. Japan, reported separately, shows essentially no differential. Country-by-country estimates and alternative treatments of cross-country dependence are reported in Internet Appendix Section IA.6.

2.6 Holding Periods

The concentration survives the holding horizons at which momentum is actually traded. Table 4 forms the overlapping $12/1/K$ strategy of Jegadeesh and Titman (1993), which holds the K most recent monthly cohorts at once, and splits its return by the calendar position of each day. With a six-month holding period, WML earns 72.6 basis points per month, 55.0 of them on the PreTOM window days ($t = 3.9$) and 17.5 on all other days. The loser leg splits the same way: −45.8 basis points per month relative to the market, −43.2 on PreTOM window days and −2.6 on the others.

2.7 Evolution over Time

Momentum weakens after 2002, but the decline occurs outside the PreTOM window.[14] Splitting the sample at the end of 2001, which divides the 552 months into 264 and 288, the rest-of-month WML rate falls from 7.01 basis points per day to −1.95, a change of −8.96 ($t = -2.38$), while the PreTOM window rate is essentially unchanged, 10.24 against 10.06 (Table 5). Over 2002–2025 the PreTOM window contributes +60.4 basis points per month and the other days −28.7 (Internet Appendix Table IA.26). The momentum premium that

[13] The differential is not market exposure. Adjusting each country's losers by their own estimated beta on the local market, which averages 1.05, leaves 2.89 basis points ($t = -2.09$), negative in 14 of the 19 markets.

[14] McLean and Pontiff (2016) and Chordia et al. (2014) document a broad decline in anomaly returns, the latter following decimalization in 2001; Bhattacharya et al. (2017) document the post-2000 attenuation of U.S. momentum specifically.

survives after 2002 is therefore entirely earned in the PreTOM window.[15]

A potential concern is that these patterns reflect changes in the sample rather than in trading: NASDAQ stocks enter CRSP in 1972, and the universe shifts toward smaller firms over time. Restricting every era to NYSE-listed stocks removes this composition effect, and the pattern holds. After 2002, the loser leg loses 33.6 basis points per event month. The six PreTOM window days account for 40.9 of that loss and the remaining days contribute +7.3, so the PreTOM window again accounts for more than the entire loss. Losers underperform the market across the whole month early in the record, so the PreTOM differential emerges only gradually. On NYSE stocks, the loser PreTOM-minus-Rest differential is +0.54 basis points per day in 1927–1951, −1.69 in 1952–1961, −4.73 in 1962–1979, −5.10 in 1980–2001 and −7.06 after 2002. Before 1952 the loser pressure is present but sits closer to month-end. Of the six-day blocks ending between τ and $\tau-14$, the one ending at $\tau-2$ shows the largest loser differential over 1927–1951, −10.66 basis points per day ($t = -2.94$). In 1952–1961 no block shows one.

The PreTOM loss build-up runs alongside the rise in institutional ownership, which reaches 67% by 2010 (Blume and Keim, 2012). Active equity funds have also cut their cash buffers, from a median of 4.5 percent of assets in 1992–1997 to 1.1 percent in 2016–2024 (Internet Appendix Section IA.8). Chernenko and Sunderam (2016, 2020) find that such buffers are too small to keep fund flows from moving prices. The increase in the PreTOM effect across eras is descriptive evidence consistent with an institutional source to the patterns that we document. Internet Appendix Section IA.7 reports the subperiod evidence.

3 The PreTOM Window's Contribution to Momentum

Do the PreTOM losses reverse afterwards, and how much more momentum is earned in PreTOM than on ordinary trading days? Because the concentration is mainly loser-side, we first ask whether the losers that are sold the most subsequently recover. We then measure how much the PreTOM window adds to the momentum premium.

[15] After 2001 the rest-of-month rate falls by 8.96 basis points per day and the PreTOM rate by 0.18, but the gap between these two changes, 8.78 basis points (standard error 6.70), is not statistically significant.

3.1 Do the PreTOM Window Losses Reverse?

The loser portfolio partly rebounds after PreTOM. Table 7 cumulates the market-adjusted return of each momentum decile from the first day of the PreTOM window, following the same stocks past month-end. Having lost 47.9 basis points relative to the market by the time the PreTOM window closes, the loser decile recovers 14.2 of them at the peak of the turn-of-the-month rebound, roughly 30 percent of the initial loss. However, the rebound is temporary.[16] By τ+9 nearly all of the loss in the loser decile has reappeared. The middle deciles by contrast are flat throughout, and winners rise steadily with no reversal. Over longer horizons, stocks that newly enter the loser decile keep underperforming for about 4 months after the window and stay below their pre-window level for about 9 months (Internet Appendix Figure IA.6).

Figure 4 traces the first four months of that path for stocks that newly enter the loser decile. Over the four months after entry, their market-adjusted return cumulates to −119 basis points on PreTOM days ($t = -3.1$) and to +41 on the other days ($t = 0.6$), so the path falls in steps, one per PreTOM window, and is flat or rising in between. The daily return on PreTOM days minus that on the other days is −5.6 basis points ($t = -2.7$). New entrants to the winner decile rise steadily with no such cycle (a difference between PreTOM days and other days of −0.2, $t = -0.2$).

These reversal paths contain return components common to losers that are sold for month-end liquidity and to those that are not. For example, in the short run, the turn-of-the-month buying documented by Ogden (1990) and Etula et al. (2020) lifts all stocks, and particularly the losers, and this can overstate the reversal of selling-related losses. To link recovery to how heavily each loser stock was sold, we ask whether losers that face more PreTOM selling also recover more afterwards, among losers of the same month, as they should if the rebound reflects recovery from the short-term price pressure of that selling. We follow the same loser stocks after the PreTOM window and relate both their PreTOM loss and their subsequent returns to incremental PreTOM selling, measured as the stock's seller-initiated imbalance during the PreTOM window relative to the preceding mid-month days.

[16] If evaluated against the losers' own returns outside PreTOM and the turn of the month, the rebound looks larger. That comparison is, however, biased toward recovery, because those days already carry part of the month-end selling (Internet Appendix Section IA.2).

Formally, for loser stock i in monthly cohort m we estimate

$$\mathrm{CAR}_{i,m}(h) = \alpha_m + \beta_h S_{i,m} + \gamma' X_{i,m} + \varepsilon_{i,m}, \tag{3}$$

where $\mathrm{CAR}_{i,m}(h)$ is the cumulative market-adjusted return accumulated from the first day of the PreTOM window through horizon h, and $S_{i,m}$ is incremental PreTOM selling, the stock's seller-initiated imbalance during the PreTOM window minus its rate over the preceding mid-month days. The cohort effect α_m absorbs everything common to the loser stocks of a given month, and $X_{i,m}$ collects predetermined characteristics measured before the PreTOM window opens, namely momentum rank, a formation-period return measure, size, return volatility and market beta over the preceding year, log bid-ask spread, log dollar volume and the mid-month selling rate itself. Standard errors are two-way clustered by stock and by the calendar month in which the horizon ends. Each horizon is estimated separately on the same observations.

Figure 5 traces the cumulative return associated with incremental PreTOM selling. A one-standard-deviation increase in incremental selling is associated with a PreTOM window loss of 131.6 basis points (standard error 9.9). Over the seven trading days that follow PreTOM that gap closes by 8.7, so only about 7 percent of the cross-sectional loss associated with higher PreTOM selling reverses promptly and 93 percent remains. The 95 percent confidence interval for that share runs from -11 to 24 percent, and full recovery within the next seven days is firmly rejected, here and in the unweighted and restricted samples (Internet Appendix Table IA.5).

Over longer horizons the value-weighted estimates of price pressure lose precision. Panel B of Figure 5 extends the estimated price response through twelve months. The estimated gap widens through month three and then narrows toward zero, but the confidence intervals widen with the horizon. Under equal weighting, 211.6 of the initial 305.1 basis points are still outstanding at twelve months, and every interval excludes zero (Internet Appendix Figure IA.5).[17]

[17] A direct return-on-return specification gives a similar short-run estimate for the rebound (Internet Appendix Section IA.9). Alternative specifications and international evidence are in Internet Appendix Tables IA.24 and IA.25. Beyond the first month after month-end, the returns that follow a PreTOM loss are statistically indistinguishable from those that follow a loss over an ordinary six-day block earlier in the same month (Internet Appendix Table IA.6).

3.2 The PreTOM Window's Contribution to the Monthly Momentum Premium

Because the losers sold hardest regain little of their PreTOM losses (Section 3.1), the window's excess return over ordinary days adds to the monthly momentum premium rather than shifting it within the month. PreTOM contains 29% of trading days and earns 63% of the monthly momentum premium. That raw share may overstate the PreTOM window's contribution to that premium if momentum would have earned positive returns on those six days anyway. The relevant counterfactual is what the same portfolio would have earned on those days under ordinary conditions. Because that return is unobservable, we benchmark PreTOM against the same fixed monthly portfolio on days outside the PreTOM window:

$$\text{Benchmarked PreTOM contribution} = 6 \times \Big(\text{PreTOM window daily return} - \text{benchmark daily return}\Big).$$

Dividing this benchmarked PreTOM contribution by the average monthly WML premium gives the benchmarked share of momentum.

Fixed monthly membership means the benchmarked contribution compares the same stocks across two parts of the same month. It tells us how much more momentum earns in the PreTOM window than on other days of the same month.

We use two benchmarks. Mid, $[\tau'+4, \tau-10]$, covers the trading days immediately preceding PreTOM. It stays inside the same monthly momentum assignment and avoids the turn-of-month days, which may themselves carry unusual returns, but because losers begin to underperform around $\tau'+8$ (Section 2.3), it is likely to reflect some of the price pressure from month-end liquidity selling. Rest uses every day outside PreTOM.

Both benchmarks give nearly the same answer, as do the alternative benchmarks of Internet Appendix Section IA.22. The PreTOM window contributes about 45.3 to 46.6 basis points per month above the benchmark days, roughly half the average monthly momentum premium (Table 6, 1980–2025). Against Mid, the abnormal return is 45.3 basis points, or 47% of the momentum premium, and against Rest 46.6, or 48%, with t-statistics of 2.1 and 2.2 from paired monthly contrasts.

The loser leg supplies most of the benchmarked contribution.[18] Its size depends more on the benchmark than WML's does, because losers rebound around the turn of the month, and Rest includes those rebound days, which raise the losers' benchmark return. Against Rest, losers account for 42.7 of the 46.6 basis points of abnormal WML ($t = 2.7$). Against Mid, they account for 30.3 of 45.3 ($t = 1.9$), a smaller and less precise share, as expected from a benchmark that already contains part of the selling.[19]

Two features bias the benchmarked contribution toward zero. First, the losers' contribution is measured relative to the market, and the market's own decline during PreTOM reflects the same month-end selling (Etula et al., 2020), so subtracting it removes part of the losers' decline.[20] Second, liquidity-driven selling of losers also occurs outside the PreTOM window, beginning around $\tau'+8$, which lowers the benchmark returns and narrows the measured contrast. The Mid benchmark contains this earlier selling and none of the turn-of-the-month rebound, so it gives the lower of the two estimates. The benchmarked contribution is therefore a lower bound, and the raw share of 63 percent an upper bound.

Finally, the estimates are not linked to momentum crashes. Excluding the 2001–02, 2009 and 2020 crash episodes leaves the estimates essentially unchanged and raises their precision, because crashes add variance that is concentrated at the start of the month (Table 6, Panel B).

3.3 Momentum Crashes

Momentum crash days are about as frequent in the PreTOM window as its share of the calendar implies, and the positive PreTOM window mean persists, and in fact rises, when its worst days for momentum are excluded (Internet Appendix Section IA.4). Crashes instead cluster at the start of the month, the first four trading days $[\tau'+1, \tau'+4]$, which have 32 percent more crash days than its share of the calendar (Figure 3, Internet Appendix Section IA.5). The PreTOM window abnormal return survives dropping the twenty most

[18] This refers to the PreTOM window's abnormal return relative to the benchmark days, not to how the full-month momentum premium divides between the two legs.

[19] Adjusting the losers also for their trailing one-year market beta leaves a contribution of 30.2 basis points against Rest ($t = 2.1$) and 15.7 against Mid ($t = 1.1$; Internet Appendix Table IA.12). A trailing two-year beta gives 29.9 ($t = 2.0$) and 15.8 ($t = 1.1$) (Internet Appendix Table IA.29).

[20] Losers make up about 3.5 percent of market capitalization, so the market also contains part of their own decline. Against a value-weighted market that excludes them, the losers' contribution is 43.8 basis points against Rest ($t = 2.7$) and 31.2 against Mid ($t = 1.9$).

extreme WML days, the 1% tails, and the 2001–02, 2009 and 2020 momentum crashes. For each crash we drop every month of the episode, not just its worst months (Internet Appendix Table IA.21). Progressively trimming the worst days from the window leaves its mean intact (Internet Appendix Section IA.4).

4 The Settlement Clock

Settlement reforms provide a direct test of the cash-demand mechanism. If investors sell losers to raise settled cash by month-end, shortening settlement by one day should move the trailing edge of the PreTOM window one trading day later. Under T+2, the U.S. trailing edge sits at $\tau-4$ (Section 2.3). A sale on $\tau-2$ would still settle by month-end, but internal funding deadlines, custodian processing and the gradual execution of large orders put the most concentrated selling a few days ahead of the formal deadline. If these operational lags stay the same across the reform, a one-day shorter settlement cycle moves the trailing edge by one day.[21]

The prediction concerns the old trailing edge relative to the new one rather than either day in isolation: comparing two adjacent days in the same month removes anything common to both, such as market news or a general change in loser returns after the reform, and leaves only the shift that the reform predicts. The U.S. transition from T+2 to T+1 predicts an increase in the $\tau-4$ minus $\tau-3$ return contrast as selling pressure shifts from $\tau-4$ to $\tau-3$. In Europe, which moved from T+3 to T+2 in October 2014, the edge starts one day earlier before the transition, so we expect an increase in the $\tau-5$ minus $\tau-4$ return contrast. The direction follows from the settlement logic rather than from the data.

4.1 The U.S. T+2-to-T+1 Transition

On May 28, 2024, U.S. equity settlement shortened from T+2 to T+1. The mechanism predicts that the trailing edge moves from $\tau-4$ to $\tau-3$. The loser return on $\tau-4$ improves by 58 basis points after the reform, with a standard error of 21, and the return on $\tau-3$ worsens by 28 (23). Estimated return changes on the remaining trading days are small and

[21] We place the trailing edge at $\tau-4$ based on Etula et al. (2020) and on the fact that under T+2 loser price pressure in the U.S. peaks there. Selling of losers does, however, continue until $\tau-1$ (Section 2.3).

mixed in sign. The boundary contrast, the change on $\tau-4$ minus the change on $\tau-3$, is 86 basis points with a standard error of 34 (Figure 6).[22] Nineteen post-reform months is a short sample, so the magnitude is estimated loosely, with a 95 percent confidence interval of roughly 20 to 150 basis points. Even so, the shift is statistically significant ($t = 2.55$) and goes in exactly the direction the settlement change predicts in advance.

Table 8 reports the underlying means and the regression estimates. Panel A shows the loser mean on $\tau-4$ moving from -6.90 basis points before the reform to 51.36 after, and on $\tau-3$ from -3.69 to -31.36, while the control days, trading days 5 to 8 of the same month, move only modestly. Panel B estimates the difference-in-differences regression

$$\begin{aligned} r_t^{D1} - r_t^{mkt} = \alpha + \beta_{\tau-4}\,\mathbf{1}_{[\tau-4]} + \beta_{\tau-3}\,\mathbf{1}_{[\tau-3]} + \pi\,\mathrm{T1Reform}_t \\ + \theta_{\tau-4}\,\mathbf{1}_{[\tau-4]} \times \mathrm{T1Reform}_t + \theta_{\tau-3}\,\mathbf{1}_{[\tau-3]} \times \mathrm{T1Reform}_t + \varepsilon_t, \end{aligned} \tag{4}$$

with those control days as the omitted reference, where $\tilde{r}_t^{D1} = r_t^{D1} - r_t^{mkt}$ is the market-adjusted loser return. Each θ is the change in the loser return on that day after the reform, relative to the change on the control days. Because $\theta_{\tau-4}$ and $\theta_{\tau-3}$ are both measured against the same control days, the control cancels from their difference. We therefore estimate that contrast using the monthly spread $\tilde{r}^{D1}_{\tau-4,m} - \tilde{r}^{D1}_{\tau-3,m}$, one observation per month. This spread is +85.93 basis points ($t = 2.55$) when using all months since 1980 as the pre-reform period and +85.66 ($t = 2.16$) when using only the T+2 era.

A stock-level triple difference against winners confirms that the shift is specific to losers. Panel C estimates

$$\begin{aligned} r_{i,t} - r_t^{mkt} = \theta_{\tau-4}\,\mathrm{Loser}_{i,t} \times \mathbf{1}_{[\tau-4]} \times \mathrm{T1Reform}_t + \theta_{\tau-3}\,\mathrm{Loser}_{i,t} \times \mathbf{1}_{[\tau-3]} \times \mathrm{T1Reform}_t \\ + \alpha\,\mathrm{Loser}_{i,t} + \beta_{\tau-4}\,\mathrm{Loser}_{i,t} \times \mathbf{1}_{[\tau-4]} + \beta_{\tau-3}\,\mathrm{Loser}_{i,t} \times \mathbf{1}_{[\tau-3]} \\ + \lambda\,\mathrm{Loser}_{i,t} \times \mathrm{T1Reform}_t + \mu_i + \delta_t + \varepsilon_{i,t}, \end{aligned} \tag{5}$$

where $\mathrm{Loser}_{i,t}$ is one for bottom-decile and zero for top-decile stocks, and the early-month days are the omitted reference. The boundary shift, $\theta_{\tau-4} - \theta_{\tau-3}$, is 196 basis points larger

[22] May 2024 is excluded throughout our U.S. settlement analysis, because its month-end falls on May 31 while the reform takes effect on May 28, so $\tau-4$ of that month precedes the reform and $\tau-3$ follows it. Every specification in this section therefore uses the same 19 post-reform months, and Figure 6 uses every month from 1980 to the reform and the 19 months after it, with standard errors clustered by month.

for losers than for top-decile stocks ($t = 2.20$). The gap exceeds the loser shift itself because the winner contrast moves insignificantly in the opposite direction. The result survives using $\tau-2$ instead of the early-month days as the control. An event study with twelve-month bins shows no pre-trend (eight pre-reform coefficients are jointly zero, $p = 0.64$; Internet Appendix Figure IA.1), and a placebo test comparing the non-boundary days $\tau-6$ and $\tau-7$ yields an estimate that is small and statistically indistinguishable from zero (Internet Appendix Table IA.1).[23]

A short post-reform sample raises the concern that a shift of this size could arise by chance in any 19-month period. To gauge this, we repeat the calculation at 305 placebo reform dates between 1996 and 2021, each followed by the same 19-month window. None produces a shift as large as the one after May 2024 (Internet Appendix Figure IA.2). The boundary moves one trading day later, exactly as the shorter settlement cycle predicts.

4.2 The Synchronized European T+3-to-T+2 Transition

Europe provides a similar test at a different boundary. On October 6, 2014, thirteen European markets in our sample moved from T+3 to T+2 equity settlement under the Central Securities Depositories Regulation (European Central Securities Depositories Association, 2014; International Capital Market Association, 2014; European Central Bank, 2014). European loser returns display roughly the same PreTOM window as U.S. loser returns, both over the full sample and in the pre-reform T+3 period (Internet Appendix Figure IA.8).[24] Under T+3, the last sale that settles by month-end falls on $\tau-3$, one day earlier than under T+2. If the trailing edge keeps the same two-day distance from that last sale as in the U.S., it sits at $\tau-5$. The European reform therefore predicts that $\tau-5$, the old trailing edge, improves relative to $\tau-4$, the new one.

[23] The U.S. 2017 T+3-to-T+2 transition gives the predicted sign for $\theta_{\tau-5} - \theta_{\tau-4}$ in a regression similar to equation (4), but with the trailing edge one day earlier. The change is insignificant, however (Internet Appendix Table IA.18). We conjecture that the difference arises because only the 2024 reform aligned stock settlement with the T+1 settlement of mutual fund redemptions (Securities and Exchange Commission, 2023). While the gap remained, funds had to bridge it with cash on hand or credit, which may have loosened the link between the timing of their stock sales and the settlement deadline.

[24] Etula et al. (2020) use the same reform, testing whether the market index's boundary-day return reversal moved in the treated countries relative to markets that kept T+3, and find that it did. They study market autocorrelation and we study loser returns; their test asks when the aggregate reversal occurs, ours asks which stocks bear the selling.

For each country we estimate

$$r^{D1}_{c,t} - r^{mkt}_{c,t} = \alpha_c + \beta_{c,\tau-5}\,\mathbf{1}_{[\tau-5]} + \beta_{c,\tau-4}\,\mathbf{1}_{[\tau-4]} + \pi_c\,\mathrm{T2Reform}_t + \theta_{c,\tau-5}\,\mathbf{1}_{[\tau-5]} \times \mathrm{T2Reform}_t + \theta_{c,\tau-4}\,\mathbf{1}_{[\tau-4]} \times \mathrm{T2Reform}_t + \varepsilon_{c,t}, \quad (6)$$

with a mid-month control window, $[\tau-15, \tau-10]$, as the reference, and aggregate the country contrasts $\kappa_c = \theta_{c,\tau-5} - \theta_{c,\tau-4}$ with inverse-variance weights, treating the country as the replication unit. Inverse-variance weighting gives each country's estimate a weight proportional to its precision, so markets with more stocks and less noisy returns count more. Because the thirteen markets changed settlement on the same date and their returns move together, the transition is one European event rather than thirteen independent ones. For inference we therefore also estimate the pooled panel of all thirteen markets with standard errors clustered by month.[25]

Using each country's own market as the benchmark, the pooled boundary shift across the thirteen markets is +14.73 basis points, with 10 of the 13 markets moving in the predicted direction (Table 8, Panel D). The benchmark-free WML contrast $\theta_{c,\tau-5} - \theta_{c,\tau-4}$ moves by −22.7 basis points, in the expected sign for a portfolio that is short losers.[26]

In the pooled daily panel with country and calendar-month fixed effects, clustered by calendar month, the boundary contrast is +17.65 basis points with a standard error of 8.82 ($t = 2.00$, $p = 0.046$), against a standard error of 5.61 that the inverse-variance weights imply when the countries are treated as independent. Clustering on country as well leaves the standard error almost unchanged (8.75), but with only thirteen countries the test has twelve degrees of freedom, so $p = 0.067$. The WML contrast is −25.16 basis points ($t = -1.85$). Because WML subtracts loser returns from winner returns, the common market benchmark cancels.

Europe is therefore a second setting in which the trailing edge moves in the predicted

[25]The thirteen markets are Austria, Belgium, Denmark, Finland, France, Ireland, Italy, the Netherlands, Norway, Portugal, Sweden, Switzerland and the United Kingdom. Germany is excluded because its equities already settled T+2, and Spain because it switched later, on 29 September 2016. They are used as controls over the 24 months before Spain's switch and show no boundary shift of their own (+10.67 basis points, $t = 0.33$). The triple difference against them is +28.30 basis points ($t = 1.32$).

[26]Excluding Switzerland, which changed on the same date but outside the Central Securities Depositories Regulation, gives +17.93, and no single country drives the estimate: dropping any one leaves the pooled contrast between +11.98 and +17.93 (Internet Appendix Figure IA.9).

direction in most markets. It is estimated less precisely than in the U.S., where the reform, identified within one large market, gives a sharper measurement. The response is concentrated at the settlement-implied boundary pair. At $\tau-7$ versus $\tau-6$, a pair that lies outside both the old and the new boundary, the reform should not change the return gap between them. Indeed, the estimate is -1.15 basis points per day ($p = 0.842$), with 7 of thirteen countries positive. Among all adjacent pairs from $\tau-9$ onward, estimated the same way, the settlement-implied pair is the only one significant at conventional levels. Internet Appendix Section IA.17 reports the country estimates, leave-one-country-out results, alternative covariance estimates, and the timing of the change. The two reforms move the loser return pressure one trading day closer to month-end at two different boundaries, in two different regions, exactly where the settlement clock predicts.

5 Selling Pressure and Portfolio Choice

5.1 Selling Pressure on Losers

We estimate the analogue of equation (1) with daily net selling pressure as the dependent variable on the 2003–2022 TAQ panel, 17.9 million stock-days:

$$\text{NetSell}_{i,t} = \beta_1 \,\text{Loser}_{i,t} + \beta_2 \,\text{Loser}_{i,t} \times \text{PreTOM}_t + \mu_i + \delta_t + \varepsilon_{i,t}, \tag{7}$$

where $\text{NetSell}_{i,t}$ is seller-initiated volume minus buyer-initiated volume over total volume, a variable that lies within $[-1, 1]$. Trade direction comes from the WRDS Intraday Indicators, which sign each TAQ trade with the Lee–Ready algorithm. The measure therefore captures which side initiated each trade rather than the total quantity liquidated by any particular investor class.[27]

Table 9 reports $\beta_2 = +0.20$ ($t = 2.50$), indicating that the difference in seller-initiated imbalance between losers and other stocks increases during PreTOM. Column 2 drops the stock fixed effect, which roughly triples the Loser main effect (consistent with stocks that become losers having higher seller-initiated volume in other months as well), and leaves the

[27] Lee–Ready signing infers the trade aggressor from transaction and quote data (Odders-White, 2000; Holden et al., 2023), so the estimate is an increase in inferred seller-initiated imbalance and does not reveal trader identity.

window interaction essentially unchanged.

On ordinary days, losers' net selling exceeds that of other stocks by $\beta_1 = 1.22$ percentage points of volume ($t = 20.20$), and during PreTOM this difference widens by a further 0.20 points. The sign of β_2 is the one the mechanism predicts.

Where the PreTOM loss falls within the day is also consistent with the selling pressure hypothesis. Lou et al. (2019) show that momentum returns accrue primarily overnight. If the PreTOM window loss reflects selling pressure, the incremental return should accrue while the market is open, and it does. Over 1992–2024, the years for which CRSP opening prices exist, the loser PreTOM window differential is −7.03 basis points per day from open to close ($t = -2.61$), against −1.52 from close to open ($t = -0.81$), with a full close-to-close differential of −8.76 ($t = -2.47$, Internet Appendix Table IA.10).[28] The additional loser loss during PreTOM therefore accrues during trading hours, when the selling takes place.

Da and Zhang (2025) identify a weekly component of momentum: institutions continue trading the same stocks in the same direction on the same weekday, which could generate recurring loser selling without any month-end cash demand. The two clocks are separable. Losers that performed especially poorly on a given weekday during the formation period do not perform especially poorly when that weekday recurs in the PreTOM window; and the PreTOM window loss is not concentrated on any particular weekday. Most directly, shortening settlement moves the loser-return boundary closer to month-end in both the United States and Europe (Section 4), and a weekly trading pattern has no reason to move with the settlement cycle. The two mechanisms are therefore complementary.[29]

Different motives to sell before month-end concentrate selling on particular dates, such as window dressing at quarter-ends and tax-loss selling in December, or on particular stocks, such as index rebalancing among S&P 500 constituents. The PreTOM window loss does not depend on any of these dates. It is present outside quarter-ends and December, and among both S&P 500 constituents and other stocks (Internet Appendix Tables IA.9 and IA.19). These motives may still operate in the background of the monthly loser liquidation, but they

[28] The winner leg reproduces the Lou et al. pattern on every day of the month, earning about 6.26 basis points per day overnight and losing −3.06 to −4.56 intraday in the PreTOM window and outside it, with no significant PreTOM window differential in either session.

[29] Sorting losers on their own formation-period return during the weekdays that match the observation date gives a top-minus-bottom tercile difference of −1.0 basis points ($t = -0.38$). Equality of the loser window differential across weekdays is not rejected ($p = 0.45$ on all exchanges, $p = 0.23$ on NYSE). Internet Appendix Section IA.12 reports the weekday cells and our replication of their headline share.

do not drive it.

Collectively, the timing evidence separates the PreTOM window loser stock loss from other known timing patterns in momentum. It accrues during trading hours, it is not organized by same-weekday continuation, it does not depend on quarter-end, December or S&P 500 membership, and it moves when the settlement clock changes. That last result makes a purely information-based reading of the PreTOM window loss difficult.

We complement the timing evidence against an information-based story by asking directly whether the additional selling in the PreTOM window carries information about subsequent fundamentals. We measure the news that follows in two ways, at the stock's next quarterly earnings announcement after the month-end. The first is the earnings surprise, the change in quarterly earnings per share from the same quarter a year earlier, divided by the standard deviation of that change over the previous eight quarters. The second is the stock's return minus the market return over the three days around the announcement. For each loser stock-month, we split net selling into two parts. Selling outside the PreTOM window is the stock's average daily net selling on the other days of the month. The PreTOM increment is how much higher its average daily net selling is inside the window than outside it. If the extra window selling were informed, the increment would predict a negative surprise and a negative announcement return at least as well as selling outside the PreTOM window does. We include both measures, together with calendar-month fixed effects and controls for size, turnover, book-to-market and the stock's return in the window, in regressions for each of the two outcomes. The sample is the 144,591 loser stock-months for which trade direction is observed, 2003 to 2022 (Internet Appendix Table IA.8).

The extra selling during PreTOM does not predict adverse news. Its coefficient is +0.026 ($t = 1.30$) on the next standardized earnings surprise and +2.05 ($t = 0.10$) on the three-day announcement return, both positive rather than negative and insignificant. A one-standard-deviation increase in the increment moves the next announcement return by +0.4 basis points, with a 95 percent interval of $[-6.8, +7.5]$. Selling outside the PreTOM window, by contrast, weakly predicts adverse news, with coefficients of −0.065 ($t = -1.71$) and −64.2 ($t = -1.74$), and the coefficient on the PreTOM increment differs significantly from the coefficient on selling outside the PreTOM window, for both the earnings surprise ($p = 0.004$) and the announcement return ($p = 0.042$).[30] Together with the settlement evidence, these

[30]Because the increment is a difference of two averages, it is measured more noisily than selling outside

results point away from informed selling as the source of the PreTOM window loss.

5.2 Mutual Funds' Portfolio Choice under Outflows

Losers face more outflow pressure from mutual funds than winners do. We multiply each fund's daily Morningstar flow, as a fraction of its assets, by the fraction of the stock's shares outstanding that the fund holds, and sum across funds (Section 2). Over 2008–2023, value-weighted losers face 0.76 basis points of shares outstanding more outflow pressure than winners over the six-day PreTOM window ($t = 8.22$, Figure 7), and pressure declines monotonically from losers to winners. Losers' extra outflow pressure does not arise because funds own more of the losers. It arises because the funds that own losers lose more money to redemptions. Part of these larger redemptions reflects the funds' poor past returns, which predict outflows (Coval and Stafford, 2007; Lou, 2012), but 42 percent of losers' extra outflow pressure remains after accounting for those returns.

Fund holdings reports allow us to test whether outflow pressure affects not only how much a fund sells, but also which stocks it chooses to sell. Holdings are reported quarterly, so they cannot show when within the month a fund sells. They show which positions the fund gives up when it needs cash, which is the selection half of the mechanism.

If a fund meets an outflow by scaling down its existing portfolio uniformly, losers and winners are treated alike, and the reduction in any position beyond that scaling is zero. If instead funds prefer to give up losers when facing liquidity needs, outflows should fall disproportionately on losers. This is what we expect to find: fund managers scaling down more loser positions compared to other positions at times of outflows.

We compare positions within the same fund and reporting interval, and we call a position that is absent from the next report a closure. We write $s_{f,i,r}$ for the split-adjusted shares fund f holds of stock i at report r and s_{f,i,r^+} for its holding at the next report. For every

the PreTOM window, which could push its coefficient toward zero regardless of its information content. A placebo rules this out. We build the same incremental selling measure for a six-day block in mid-month, $[\tau-15, \tau-10]$, for the same loser stock-months, from the same TAQ data, and over the same number of days, and rerun the regression with this block in place of the PreTOM window. The coefficient on the mid-month increment is -0.024 for the earnings surprise and -27.9 for the announcement return. Both are negative, as is the coefficient on selling outside the block, and the test that the two coefficients are equal is not rejected for either outcome ($p = 0.16$ for the earnings surprise and $p = 0.26$ for the announcement return). Only the PreTOM increment differs from selling on the other days of the month.

position held at report r, that is with $s_{f,i,r} > 0$,

$$\text{Trade}_{f,i,r} = \frac{s_{f,i,r^+}}{s_{f,i,r}} - 1, \qquad \text{ExcessLiq}_{f,i,r} = \phi_{f,r} - \text{Trade}_{f,i,r},$$
$$\text{Closure}_{f,i,r} = \mathbf{1}\{s_{f,i,r^+} = 0;\ s_{f,i,r} > 0\}, \tag{8}$$

where $\phi_{f,r}$ is the fund's net flow over the interval as a fraction of its assets at the start. $\text{Closure}_{f,i,r}$ is an indicator equal to one when the position is absent from the next report, and ExcessLiq denotes excess liquidation, the reduction in a position beyond what proportional scaling implies. Any proportional liquidation removes the same fraction of every position, so it leaves ExcessLiq equal across a fund's holdings. Our regressions compare positions within the same fund and report, and the null is that losers' ExcessLiq equals that of the fund's other positions.[31]

Holdings are from Thomson S12, which stamps each portfolio both with the date it was reported and with the vintage of the file it appears in. One report can appear in several later vintages, so pairing on file dates could read that a portfolio was carried forward by a quarter with no trading. We make sure to pair on report dates instead, keeping only intervals of 45 to 200 days, which spans the quarterly and semiannual reporting cycles. Fund flows are the implied flows of Coval and Stafford (2007), computed from CRSP monthly net assets and returns and cumulated over each reporting interval. The sample begins in 1991, when monthly CRSP net assets become sufficiently available to construct monthly flows,[32] and contains 49.4 million fund-stock-report observations across 11,185 funds through 2024.

We begin by asking how funds treat loser positions before conditioning on outflow pressure. A loser position is absent from the fund's next holdings report 14.3% of the time, against 8.9% for the middle deciles (Panel A of Table 10).[33] The raw closure-rate gap may partly reflect which funds hold loser and middle-decile stocks. Panel B therefore compares

[31] $\phi_{f,r}$ and $\text{Trade}_{f,i,r}$ are both proportions. A fund holding 70 percent of its assets in one stock and 30 percent in another that meets a 50 percent outflow pro rata sells half the shares of each, so the split stays at 70/30 and ExcessLiq is zero on both. It is larger on a position the fund shrank faster than its other holdings. When returns accrue during the interval, a pro rata sale removes a fraction of each position that differs from $\phi_{f,r}$, but by the same amount for every position, and the fund × report fixed effects absorb it.

[32] Before 1991 CRSP reports mutual-fund net assets predominantly quarterly or annually. The number of fund-months with a computable flow rises from 791 in 1990 to 22,127 in 1991.

[33] Both rates exclude positions in stocks that CRSP delists during the reporting interval, which would be recorded as closed whether or not the fund sold them. Before that screen they are 15.3% and 9.4%.

positions within the same fund and reporting interval, estimating the two outcomes separately,

$$Y_{f,i,r} = \beta_1 \text{Loser}_{i,r} + \beta_2 \text{Winner}_{i,r} + \alpha_{f,r} + \varepsilon_{f,i,r}, \qquad Y \in \{\text{Closure}, \text{ExcessLiq}\}, \tag{9}$$

where Y is either the closure dummy or excess liquidation, $\text{Loser}_{i,r}$ and $\text{Winner}_{i,r}$ are indicators equal to one for stocks in the bottom and top momentum deciles at report r, $\alpha_{f,r}$ is a fund $\times$ report fixed effect, and the middle momentum deciles (D2–D9) are the omitted group. Within the same fund-report, loser positions are 6.51 percentage points more likely than middle-decile positions to be absent from the next report ($t = 16.66$), and their excess liquidation is 4.53 percentage points higher ($t = 9.94$). Winners differ little from the middle deciles, at -0.18 ($t = -1.31$) and -0.91 ($t = -4.57$).[34]

Table 10 shows that funds close loser positions more often than middle-decile positions, a loser tilt in closures.[35] That estimate averages over all reporting intervals, those in which money flowed into the fund and those in which it flowed out. If the tilt reflects cash needs, it should be larger when money leaves the fund and should not respond to money arriving. Table 11 tests this by letting the tilt vary separately with outflows and inflows,

$$\begin{aligned} \text{Closure}_{f,i,r} = {} & \beta_1 \text{Loser}_{i,r} + \beta_2 \text{Winner}_{i,r} + \alpha_{f,r} \\ & + \gamma_1 \text{Loser}_{i,r} \times \text{Outflow}_{f,r} + \gamma_2 \text{Winner}_{i,r} \times \text{Outflow}_{f,r} \\ & + \delta_1 \text{Loser}_{i,r} \times \text{Inflow}_{f,r} + \delta_2 \text{Winner}_{i,r} \times \text{Inflow}_{f,r} + \varepsilon_{f,i,r}, \end{aligned} \tag{10}$$

where $\text{Outflow}_{f,r} = \max(-\phi_{f,r}, 0)$ and $\text{Inflow}_{f,r} = \max(\phi_{f,r}, 0)$ are the two sides of the flow. The tilt strengthens with net outflows but changes little with net inflows. The Loser $\times$ Outflow coefficient is 0.0728 ($t = 4.66$, column 1 of Table 11), so a one-standard-deviation increase in net outflows raises the loser-versus-middle-decile closure gap by 0.44 percentage points. Loser $\times$ Inflow is 0.0042 ($t = 0.47$). The Winner $\times$ Outflow coefficient has the

[34] In raw terms winners are closed more often than middle-decile positions (Panel A of Table 10), but within the same fund and report the difference disappears. The raw gap reflects which funds hold winners, namely funds that close more of all their positions, rather than how a fund treats its winners.

[35] Mechanical rebalancing cannot account for a loser tilt in either direction. Restoring a target weight after a stock falls would require buying the stock rather than selling it. Rules that prune positions once their weight becomes small are addressed by the position-weight controls, and the outflow response holds within every weight quintile (Table 11).

opposite sign, -0.0181 ($t = -2.49$).[36]

It is possible that loser positions tend to be smaller or otherwise different from the rest of the portfolio, so the outflow interaction could reflect which positions are easiest to eliminate rather than their momentum rank. Column 2 controls for initial portfolio weight, the bid-ask spread, firm size and dollar volume, together with their interactions with outflows. The Loser $\times$ Outflow coefficient falls to 0.0330 but remains positive ($t = 2.77$).[37] Column 3 avoids imposing a common linear relation with position size. We divide each fund's positions into quintiles by their initial portfolio weight and estimate the specification separately within each quintile, so a loser is compared with middle-decile positions in the same position-weight quintile held by the same fund at the same report. The loser-outflow interaction is positive in all five groups, 0.108 for the smallest positions and 0.046, 0.042, 0.042 and 0.035 as position size increases (column 3). Outflows hit small positions especially hard, and beyond the smallest quintile the loser tilt is nearly constant. Small position size amplifies the closure response without fully accounting for the preferential closure of losers.[38]

The outflow response is concentrated at the extensive margin, in whether a position is closed rather than in how much of it is sold. Estimating the same specification with excess liquidation in place of the closure indicator gives a loser-outflow interaction of 0.0312 ($t = 1.59$), smaller and less precisely estimated. Counting every position, closures included, funds reduce losers faster than their middle-decile holdings, by the 4.53 percentage points of Table 10.[39]

[36] Report dates are 45 to 200 days apart, so a longer interval accumulates both more outflow and more time to close a position. Adding Loser and Winner interactions with the length of the interval leaves Loser $\times$ Outflow at 0.0591 ($t = 3.83$), and measuring outflows per quarter of interval gives 0.0439 ($t = 2.78$).

[37] Column 2 also interacts the loser and winner indicators with the fund's trailing twelve-month return. Window dressing before a report date is a performance-linked motive, in which poorly performing funds sell losers so that they do not appear in the disclosed portfolio (Lakonishok et al., 1991). The trailing-return interactions absorb that motive, and the outflow response survives them. Report-date window dressing also cannot produce the return pattern, which holds in months without a quarter-end (Internet Appendix Table IA.9).

[38] Thomson S12 mutual-fund holdings have no minimum reporting size. The exemption for positions below 10,000 shares and $200,000 applies to 13F institutional filings, not to fund portfolio reports, so a closure cannot arise from a small position falling below a reporting threshold. Restricting the sample to positions worth at least $100,000 or $1 million at the start of the interval makes the Loser $\times$ Outflow coefficient larger, 0.122 ($t = 7.51$) and 0.152 ($t = 7.71$).

[39] Among positions that remain in the portfolio, losers are trimmed 3.39 percentage points less than the middle deciles relative to pro rata ($t = -13.32$). That comparison conditions on the fund keeping the position. If funds deal with unwanted losers by closing them outright, the losers that remain are by construction those the fund chose to keep, so they are trimmed less even without any reluctance to realize losses.

The clearest response to outflows is in which loser positions disappear from the portfolio. The loser asymmetry is also visible in three selected episodes of acute aggregate cash demand: September 2008, October 2008 and March 2020. Returns and fund flows are measured monthly there, so the episodes carry no information about where within the month the asymmetry falls and we report them as description rather than as a test. Momentum earns +10.5, +10.9 and +18.2 percentage points in September 2008, October 2008 and March 2020 because losers fall far more than winners. In March 2020 past-loser mutual funds also underperform their Carhart benchmark by 11.2 percentage points while past-winner funds remain near benchmark (Internet Appendix Table IA.30). In fund holdings, the loser closure tilt rises during the financial crisis. In reporting intervals ending between September and December 2008 it is 3.4 percentage points above its usual 6.5 ($t = 3.67$). In the interval ending in March 2020, by contrast, there is no detectable increase in the loser closure tilt.

The holdings evidence identifies mutual funds as one source behind selective loser selling. Funds meeting outflows do not scale their portfolios down uniformly: loser positions become disproportionately more likely to disappear, while comparable inflows produce no such response. In other words, these results show that for mutual funds at least the loser stocks are dispensable: stocks that they prefer to sell when facing liquidity needs. Within the loser decile, funds are also more likely to close positions far below their 52-week high. A one-standard-deviation increase in that distance raises the probability that a loser position is closed by 4.8 percentage points ($t = 15.1$, Internet Appendix Table IA.14), a third of the average closure rate, and by 2.9 percentage points with controls for size, liquidity and portfolio weight ($t = 11.8$). This preference holds when funds face no outflows, and with position controls it does not vary significantly with outflow size. Section 6 examines which loser stocks investors prefer to sell during PreTOM. The quarterly holdings data cannot locate sales within the month; the timing evidence comes from the daily returns, trading, and settlement changes.

6 Are Some Loser Stocks More Dispensable Than Others?

We call stocks dispensable when investors with acute cash needs prefer to sell them compared to other stocks to raise cash. We conjecture that dispensable stocks are large liquid stocks that cost little to sell, and stocks that bring their holder some kind of benefit from selling, such as a realized tax loss. We have shown evidence that cash-demand driven selling at PreTOM falls disproportionately on momentum losers, implying that loser stocks are on average dispensable. We now ask whether some loser stocks are more dispensable than others, and in particular whether stocks' distance from their 52-week high is a characteristic that identifies them.

A stock far below its 52-week high has salient past underperformance and potentially a larger and more valuable embedded tax loss compared to loser stocks in general.[40] Most such stocks also pay no dividend, so selling them gives up little income. We return to these reasons below.

Distance from the 52-week high is closely related to momentum. George and Hwang (2004) show that nearness to the 52-week high explains a substantial share of momentum profits, so an unconditional sort on it would say little about whether this measure of dispensability is distinct from the momentum rank. We therefore sort on distance from the 52-week high within the momentum deciles, which holds momentum decile membership fixed. As is the norm in the literature, we use a 12-2 sort to define the winner and loser legs. Therefore, one additional reason why distance from the trailing 52-week high can characterize dispensability within the loser leg is that it also includes the returns from the most recent month.

If the 52-week-high effect of George and Hwang (2004) sorts dispensable stocks within the loser decile, two predictions should hold. First, the return gap between losers far below their 52-week high and those near it should widen in the PreTOM window, when investors must raise cash for month-end liquidity. Second, distance from the 52-week high should predict which of the losers investors choose to sell.

Panel A of Table 12 tests the first prediction. Each month we take the stocks in the

[40] Short-term losses can be used to offset short-term capital gains, which are taxed more heavily than long-term capital gains in the U.S.

bottom momentum decile and sort only these losers into quintiles by distance from their 52-week high, so every stock in the comparison is a momentum loser. Value-weighted, the losers farthest below their high underperform those nearest it by −9.6 basis points per day in the PreTOM window ($t = -2.79$) and by −2.2 basis points outside it ($t = -0.82$). The spread thus widens by −7.3 basis points per day in the PreTOM window ($t = -1.81$, Internet Appendix Table IA.15). The 52-week-high effect of George and Hwang (2004) does not predict such widening of this spread before month-end.

Panel B tests the second prediction. Within the loser decile, net selling pressure rises monotonically with distance from the 52-week high, from 0.037 in the quintile nearest the high to 0.077 in the quintile farthest, and a one-standard-deviation increase in this distance raises selling by 0.0040 ($t = 7.89$, two-way clustered, 3.0 million loser stock-days). Selling rises with distance below the 52-week high throughout the month, while the return gap between stocks farthest below their highs and those nearest them widens during PreTOM. The two results together are consistent with a persistent asset-selection margin, investors selling dispensable stocks that are characterized by a large distance to the 52-week high, whose effects on prices intensify during PreTOM when cash needs become binding at the market level (Etula et al., 2020).

We also ask whether the 52-week-high spread in performance in the PreTOM window extends beyond losers (Internet Appendix Table IA.15). The middle momentum decile shows no spread, at −0.8 basis points per day ($t = -0.3$). The winner estimate is also indistinguishable from zero, though its point estimate is negative at −4.8 ($t = -1.6$), about half the loser magnitude. The spread is largest in the loser leg, where the mechanism places the cash-driven selling.

As discussed above, tax considerations may be one reason why stocks far below their 52-week high are dispensable. Holders who bought near the high within the past year carry a short-term loss on a stock far below the high, the most valuable kind to realize. Grinblatt and Moskowitz (2004) show that tax-loss selling contributes to momentum returns. An embedded loss raises the benefit of realizing a position whenever cash has to be raised, so it bears on which position is chosen. What our evidence rules out is that year-end tax-loss selling explains the PreTOM window selling. Excluding December, the month in which tax-loss harvesting is concentrated, leaves the losers' PreTOM window underperformance relative to the market essentially unchanged at −7.03 basis points per day against −7.15 in

the full sample (Internet Appendix Table IA.9).

Low dividends may also make stocks far below their 52-week high dispensable. Figure 8 shows how the share of non-dividend-paying stocks rises from 39.8 percent among stocks nearest to their 52-week high to 89.1 percent among those farthest below it, a much sharper increase than the conventional 12-2 momentum sort produces, where the figures are 84.0 percent of non-dividend payers among losers and 67.6 percent among winners. Investors treat dividends as a separate stream of income. They have been shown to hold high-yield stocks longer and to be less willing to sell a stock whose returns come through its dividends, and mutual funds behave the same way (Hartzmark and Solomon, 2019). Investors also spend dividends instead of selling shares when they need cash (Baker et al., 2007). A loser stock that pays no dividend provides no such income, so it is a more attractive position to give up when cash is needed.

Liquidity is a second characteristic of dispensable losers, separate from distance from the 52-week high. The PreTOM window loss is largest among losers with the narrowest prior-month bid-ask spreads, where large sales are cheapest to execute (Internet Appendix Table IA.22). Etula et al. (2020) likewise find that month-end cash needs are met by selling large, liquid stocks.

Taken together, distance from the 52-week high tells us which losers investors are more likely to sell, beyond what momentum decile membership tells us. Dispensability itself is latent, and distance from the high is only an imperfect proxy for it. We use it to rank stocks within the loser decile, not as a new return factor. Among momentum losers, distance from the 52-week high predicts both seller-initiated imbalance and the size of the PreTOM return loss. The selling hierarchy exists throughout the month, while its price consequence is concentrated in the PreTOM window.

7 Implications

7.1 Monthly Cash Demand and the Disposition Effect

The literature has suggested the disposition effect (reluctance to realize losses) as one explanation of momentum; e.g., Frazzini (2006). Our evidence points in the other direction: exceptionally large selling of losers in the PreTOM period as an explanation of momentum.

Funds facing net outflows do not hold on to their losers. Loser positions are disproportionately more likely to be absent from the next holdings report, and across all positions funds reduce losers faster than their other holdings (Section 5.2).[41] Funds holding past losers tend to have performed poorly and they face outflows (Coval and Stafford, 2007; Lou, 2012). Because of this, losers are sold more in aggregate even if managers were to scale down uniformly. Morningstar daily mutual fund flow data confirm that momentum losers face higher outflow pressure, also during PreTOM (Section 5.2). Our within-fund comparisons remove that channel by construction, so the selection effect to liquidate losers at times of outflows documented in Section 5.2 is in addition to it.[42]

7.2 Fund Performance Persistence

The intramonth momentum cycle in stocks also changes how momentum-related fund underperformance is distributed within the month. Carhart (1997) attributes most fund-performance persistence to momentum loadings and expense ratios. We find that the two components follow different calendars. Momentum exposure generates concentrated losses during PreTOM, while fees accrue throughout the month.

Kenneth French's published daily momentum factor shows the same cycle, although it is rebalanced daily and built on a different universe. It earns 5.51 basis points per day in the PreTOM window against 1.36 on the other days of the month. Our results are therefore not specific to how we construct the momentum portfolios.

We find that a fund with a negative momentum loading also faces a larger daily return drag during the PreTOM window.

For the funds with the worst past performance, the returns of our momentum loser portfolio explain their PreTOM underperformance, but do not explain their performance

[41] One potential explanation is that fund managers view outflows as evidence of a lack of trust in the funds' current investments in loser stocks, and it is for this reason that at times of outflows momentum losers are cleaned out of the funds' portfolios.

[42] An exception to the disposition effect occurs annually. Odean (1998) documents that the disposition effect reverses in December, when tax incentives make losses worth realizing. This override is by retail investors and at an annual frequency, whereas the loser-directed selling documented here recurs around each month-end. Institutional selling also departs from the disposition effect in another way. Akepanidtaworn et al. (2023) document a related salience pattern in institutional selling, where portfolio managers disproportionately sell the positions with the most extreme returns since purchase, winners and losers alike. Their reference point is the purchase price and ours is the trailing 52-week high, so the two measures are related without being the same.

outside the PreTOM window. These funds' Fama–French three-factor alpha in the PreTOM window is −2.02 basis points per day, and it rises to −0.38 once we include the loser portfolio's return in the regression explaining fund returns. Outside the PreTOM window, including the loser portfolio's return moves alpha only from −2.40 to −1.82. Adding fund-level expense ratios to the regression leaves no detectable alpha in either part of the month.

The May 2024 settlement reform moves Kenneth French's momentum factor return and the past-loser fund portfolio return one trading day later, as it moves the loser stock returns. This test uses daily fund net asset values rather than individual stock returns. It extends the settlement evidence to a different dataset and unit of observation. The factor's intra-month profile, the within-month underperformance of past-loser funds, and the separation of expense ratios from the momentum component are in Internet Appendix Sections IA.24.1, IA.24.2 and IA.24.3.

8 Conclusion

We document that U.S. equity momentum returns are concentrated in a six-trading-day window ending four trading days before month-end. This pre-turn-of-the-month window (PreTOM) accounts for 63% of the momentum premium over 1980–2025 and the entire premium since 2002, the midpoint of our sample.

The return concentration is driven by loser stocks, which experience disproportionate losses and selling pressure during PreTOM. Trade-level data show that seller-initiated order imbalances increase among loser stocks during PreTOM. Daily mutual fund flows show that loser stocks face the largest outflow pressure during PreTOM among different momentum decile portfolios. Beyond the flow driven selling of losers at PreTOM, there is a selection effect: mutual funds facing outflows are shown to prefer to disproportionately liquidate their loser positions.

Settlement reforms link the loser stocks' PreTOM underperformance to the month-end liquidity needs: the U.S. shift from T+2 to T+1 and the European shift from T+3 to T+2 each move loser-stock price pressure one trading day closer to month-end. The PreTOM pattern is also present in 19 developed markets.

The price pressures that emerge during PreTOM are not quickly reversed. Although loser stocks rebound by approximately 30% of their PreTOM losses by τ+4, the fourth trading

day of the following month, this recovery is gone by τ+9 (Table 7). New entrants to the momentum loser portfolio continue to experience losses across several successive PreTOM periods and, on average, recover their initial losses only after 9 months. Stocks subject to the heaviest selling during PreTOM fare below average: they experience larger initial PreTOM losses, rebound by only 7% of their PreTOM losses, and their prices remain depressed for months. This implies that the results we document are not merely about timing of momentum returns within the month, but about the origins of momentum returns in the modern era.

The concept of dispensability explains which of the loser stocks investors choose to liquidate and which of the loser stocks underperform during PreTOM. Among momentum losers, selling and underperformance fall disproportionately on stocks far below their 52-week high. Such stocks are particularly attractive to sell when cash is needed: they are salient underperformers, likely to offer attractive short-term losses for tax purposes, and pay little dividends.

These documented PreTOM patterns are a feature of the institutional era. Return concentration in the PreTOM window is absent during the first half of the twentieth century. It emerges gradually from mid-century onward, alongside the rise in institutional ownership. One reason for this can be that over time mutual funds, seeking efficiency, have on average reduced their cash holdings, forcing them to meet recurring month-end liquidity needs commonly through equity sales during PreTOM.

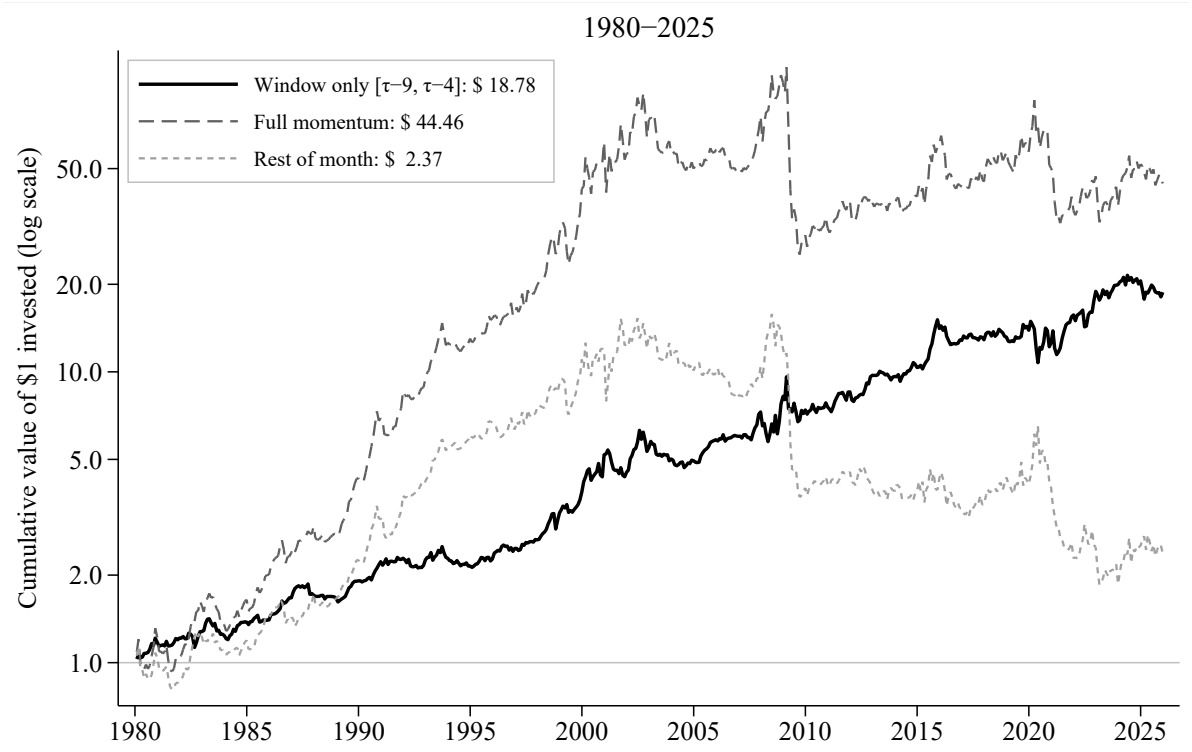


Figure 1: Cumulative wealth from $1 invested in three momentum strategies: WML earned only during PreTOM ($[\tau-9, \tau-4]$), WML earned only during Rest (the approximately 15 trading days outside the PreTOM window), and WML earned every day. WML is a self-financing long-short portfolio; on days outside the relevant window the strategy holds cash at zero return. Trading days are indexed relative to the turn-of-the-month (month-end) denoted by τ. Daily returns are value-weighted, 1980–2025. Deciles use NYSE breakpoints and are held fixed within each calendar month.

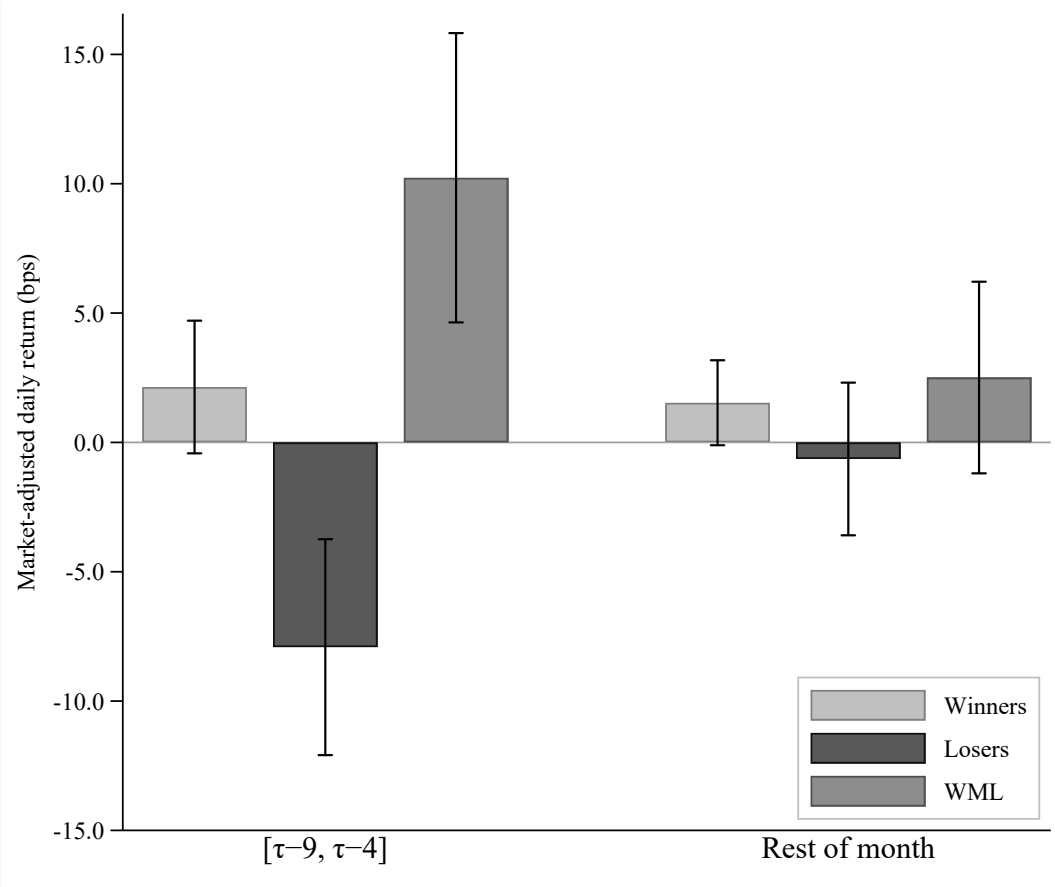


Figure 2: Average daily market-adjusted returns (basis points) of momentum winners, losers, and WML during PreTOM ($[\tau-9, \tau-4]$) versus Rest. Daily returns are compounded within each window and averaged across months; 95% confidence intervals use heteroskedasticity-robust standard errors. Daily returns are value-weighted, 1980–2025. Deciles use NYSE breakpoints and are held fixed within each calendar month.

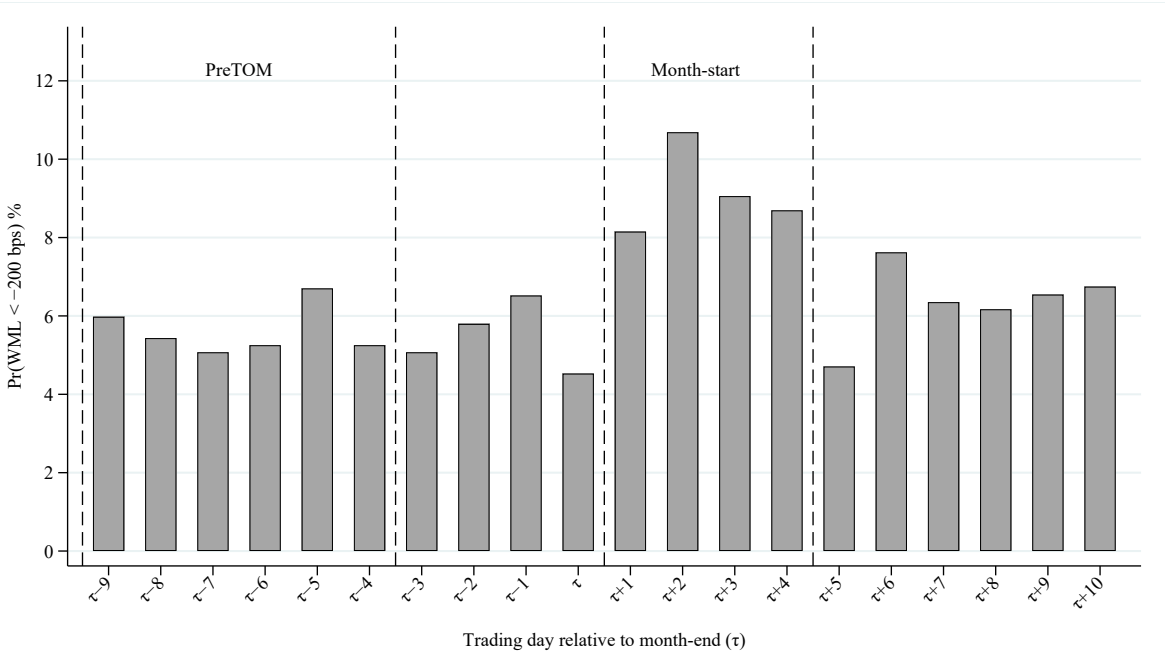


Figure 3: Crash probability by trading day relative to month-end τ, defined as Pr(WML $<$ -200 bps). Dashed vertical lines mark the PreTOM window ($[\tau-9, \tau-4]$) and the month-start window ($[\tau+1, \tau+4]$). Daily returns are value-weighted, 1980–2025. Deciles use NYSE breakpoints and are held fixed within each calendar month. The days after τ are the first trading days of a month; because deciles are re-sorted at each month-end, they are plotted after τ to show the turn of the month in sequence.

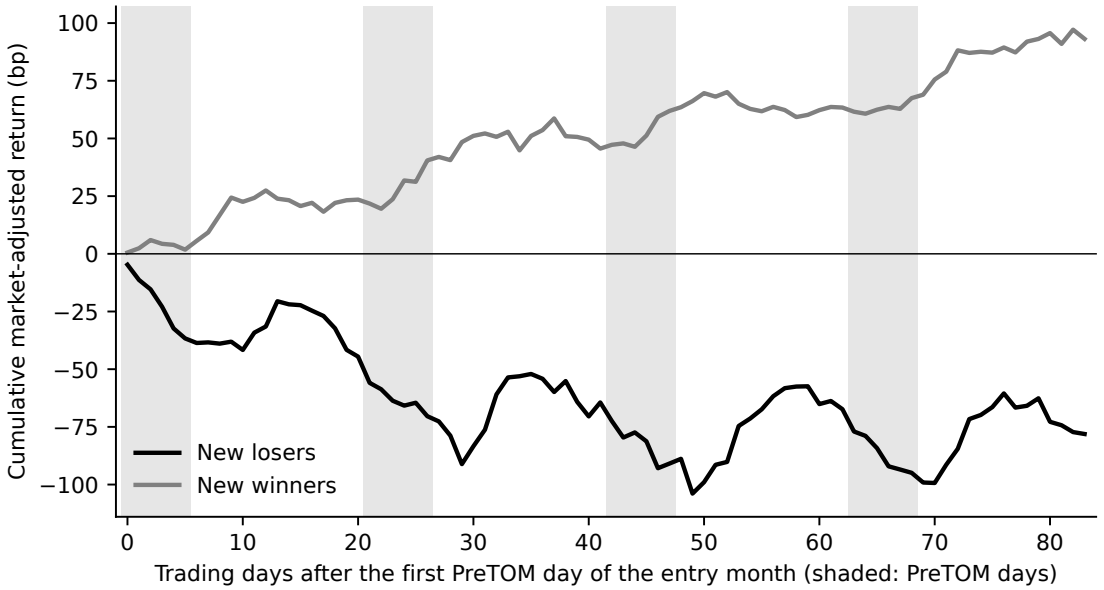


Figure 4: Cumulative market-adjusted return of new entrants to the loser decile (black) and to the winner decile (gray) over the first four months after entry, in basis points. Day 0 is the first PreTOM day, $\tau-9$, of the month in which the stock enters the decile. Shaded days are PreTOM days for most cohorts. Value-weighted cohorts with weights fixed at entry, followed whatever their later decile, CRSP delisting returns included, 1980–2025.

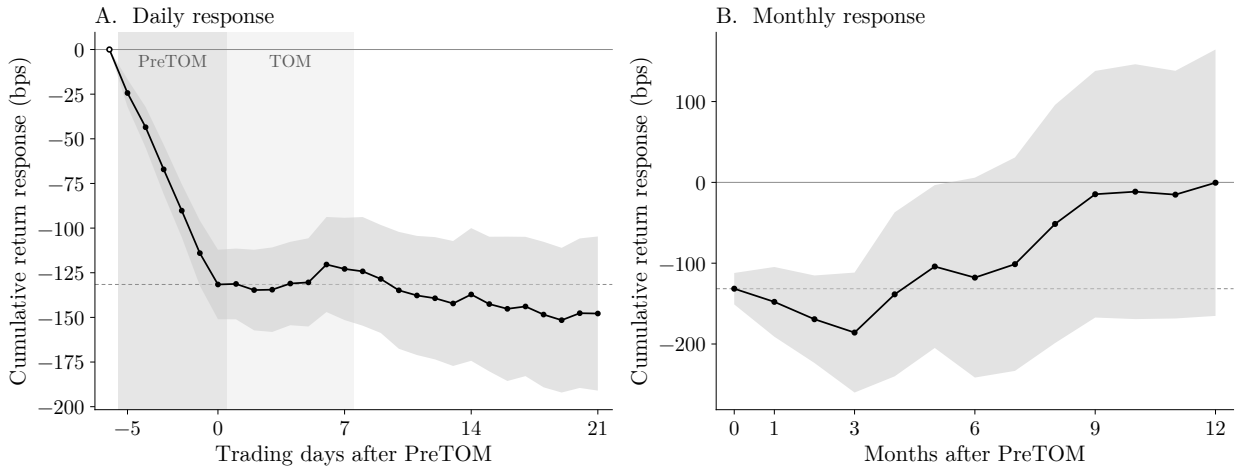


Figure 5: Cumulative average market-adjusted returns from a one-standard-deviation above normal PreTOM selling of loser stocks. Day 0 is the final PreTOM trading day. Panel A follows daily returns through 21 trading days and Panel B monthly returns through twelve months. The dashed line marks the estimated return response during PreTOM, −131.6 basis points, and zero marks full recovery. Capitalization weighted, 132,637 loser stock-months, 2003–2022, with 95 percent confidence bands based on two-way clustered (stock and outcome month) standard errors. The full specification and the recovery tests are in Internet Appendix Table IA.5.

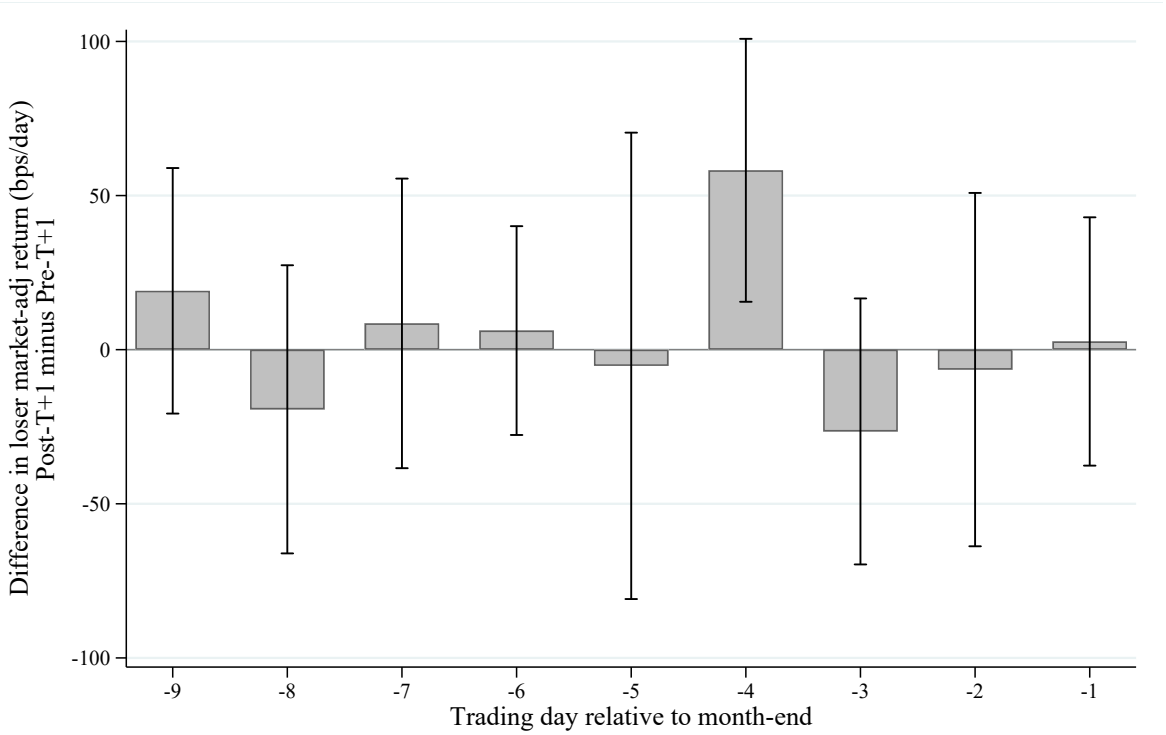


Figure 6: Difference in mean daily market-adjusted VW loser returns (Post-T+1 minus Pre-T+1) at each trading day relative to month-end ($\tau-9$ through $\tau-1$), with 95% confidence intervals. The SEC transitioned from T+2 to T+1 settlement on May 28, 2024. All bars report the difference in Post-T+1 and Pre-T+1 average market-adjusted returns. The loser portfolio is defined as in Section 2. Sample: every month from 1980 to the reform and the 19 months after it, May 2024 excluded, with standard errors clustered by month.

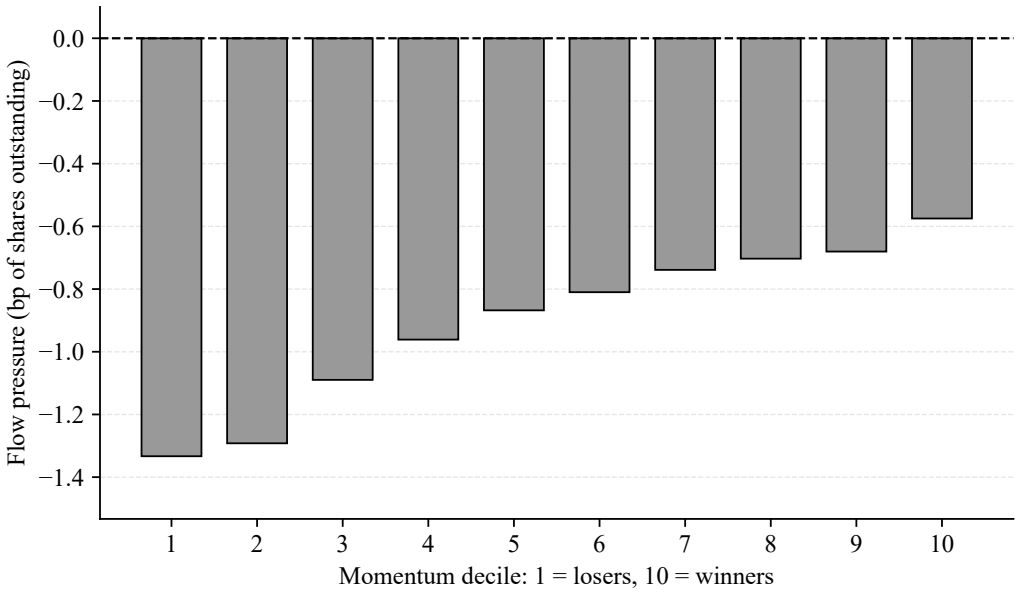


Figure 7: Mutual-fund flow pressure by momentum decile over the six-day PreTOM window ($[\tau-9, \tau-4]$), in basis points of shares outstanding, 2008–2023. Each fund's daily Morningstar flow is allocated to its holdings in proportion to their size, summed across funds, and divided by the stock's shares outstanding. Stocks are value-weighted within each decile by market capitalization at the start of the month, and each month counts equally. U.S. equity open-end funds; funds with an exchange-traded share class are excluded. Negative values are net outflows.

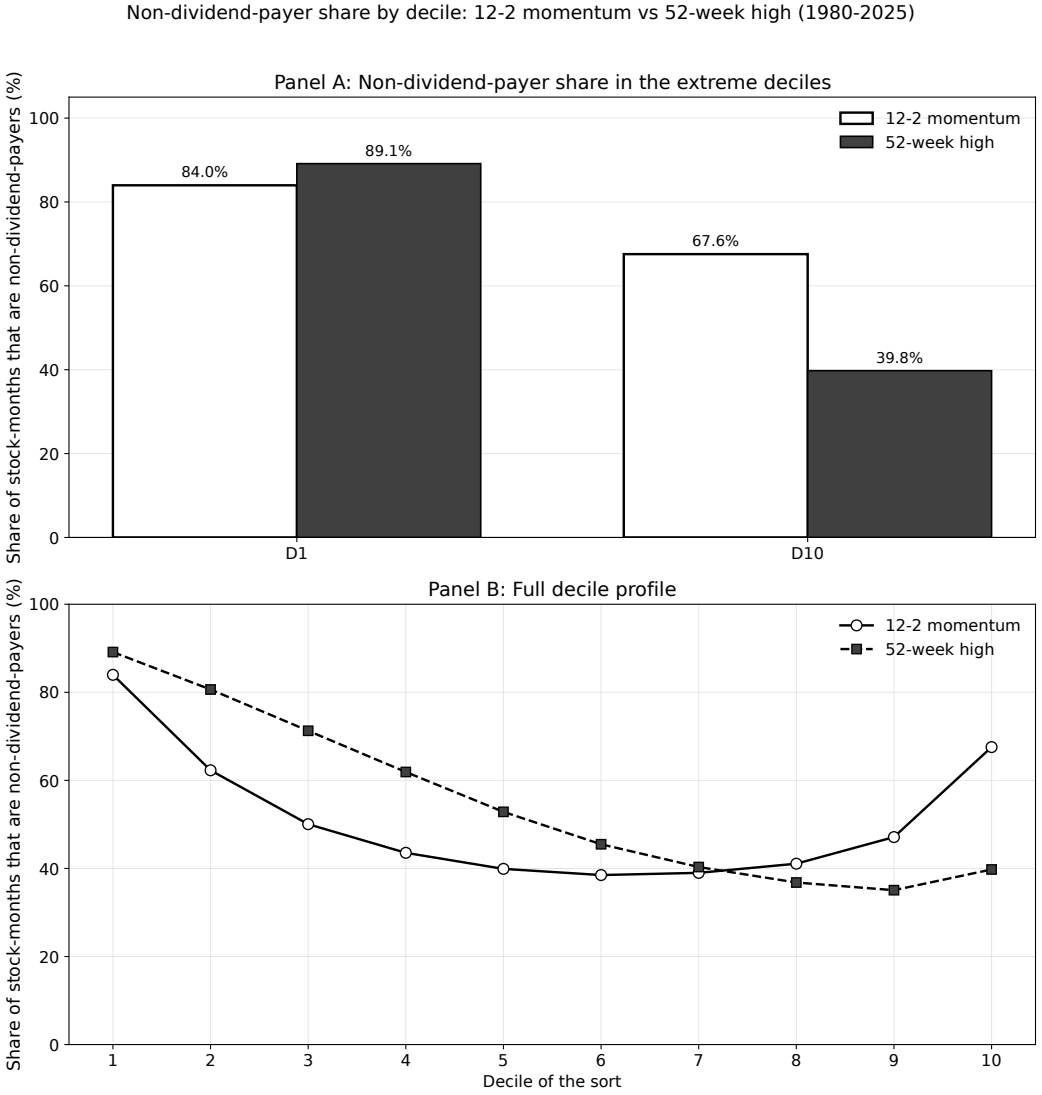


Figure 8: Share of non-dividend-paying stock-months by momentum decile under two sorts. Panel A: the two extreme deciles shown as grouped bars, with 12-2 momentum (white bars with black border) and 52-week-high distance (dark gray bars). The decile label means different things under the two sorts: under 12-2 momentum D1 holds past losers and D10 past winners, while under the 52-week-high sort D1 holds the stocks farthest below their trailing high and D10 those nearest it. Panel B: full decile profile, 12-2 momentum (solid line with hollow markers) and 52-week-high distance (dashed line with filled markers). Both sorts are formed at the end of each month from all NYSE, AMEX, and NASDAQ common stocks in CRSP; assignments are held constant within the calendar month. Dividend-payer status is taken from CRSP distributions. Sample: 1980–2025.

Table 1: Momentum Returns in the PreTOM Window and on Rest Days

	Rest	PreTOM	PreTOM − Rest
Losers (D1)	−0.87	−7.98***	−7.12***
	(−0.58)	(−3.75)	(−2.79)
Winners (D10)	1.52*	2.17	0.65
	(1.78)	(1.63)	(0.39)
WML	2.39	10.15***	7.76**
	(1.24)	(3.62)	(2.28)

Notes: Daily value-weighted portfolio returns, 1980–2025, in basis points. Losers and winners are the bottom and top momentum deciles; WML is winners minus losers. Individual-leg returns are adjusted by Kenneth French's value-weighted market return. Deciles use NYSE breakpoints and are held fixed within each calendar month. Each series is regressed on a constant and a PreTOM indicator for days $[\tau-9, \tau-4]$; Rest comprises all other days. Parentheses report Newey–West t-statistics with 21 lags. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Stock-Level Panel Regressions of Loser Returns in the PreTOM Window

	Value-weighted			Equal-weighted		
	(1)	(2)	(3)	(4)	(5)	(6)
Loser (β_1)	−1.08	2.61**	2.96*	5.11***	8.02***	0.74
	(−0.84)	(2.04)	(1.85)	(6.33)	(11.19)	(0.89)
Loser × PreTOM (β_2)	−7.08***	−7.15***	−8.04***	−2.51*	−2.55*	−4.35***
	(−3.05)	(−3.08)	(−2.82)	(−1.81)	(−1.85)	(−2.83)
Observations	53,324,149	53,324,145	45,987,241	53,324,149	53,324,145	45,987,241
Stock fixed effects	No	Yes	Yes	No	Yes	Yes
Day fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	No	Yes	No	No	Yes

Notes: Estimates of equation (1) on the full CRSP panel of NYSE, AMEX, and NASDAQ stocks, 1980–2025. The dependent variable is the stock's daily return in excess of the daily value-weighted market return from Kenneth French's data library, in basis points. $\text{Loser}_{i,t}$ equals one if stock i is in the bottom momentum decile on day t and PreTOM_t equals one on the six trading days $[\tau-9, \tau-4]$ before month-end. Columns 1 and 4 include day fixed effects only; the remaining columns add stock fixed effects. Value-weighted columns weight observations by the previous trading day's market capitalization within decile-day. Columns 2 and 5 report 53,324,145 observations rather than 53,324,149; the difference is four observations dropped as singleton groups during fixed-effect estimation. The control in columns 3 and 6 is the log of the stock's mean daily quoted bid-ask spread over calendar month $m-1$, centered on its sample mean. The regression includes it both on its own and interacted with $\text{Loser}_{i,t}$ and PreTOM_t. Requiring at least ten valid spread days in month $m-1$ restricts the sample to 45,987,241 stock-days. t-statistics in parentheses; standard errors two-way clustered by stock and day. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: International Evidence on PreTOM Returns across 19 Developed Markets, Pooled Regression, 1991–February 2026

	PreTOM	Rest of Month	Difference
Losers − Market	-5.52^{***}	-2.34^{**}	-3.18^{**}
	(-4.24)	(-2.46)	(-2.24)
Winners − Market	$+4.40^{***}$	$+2.31^{***}$	$+2.09$
	(3.84)	(3.26)	(1.56)
WML	$+10.02^{***}$	$+4.50^{***}$	$+5.54^{**}$
	(4.84)	(3.21)	(2.36)

Notes: Daily value-weighted returns across nineteen developed markets, 1991–February 2026, in basis points. Returns are computed in U.S. dollars; loser and winner returns are adjusted by each country's own market return. WML is winners minus losers. PreTOM comprises local trading days $[\tau-9, \tau-4]$; Rest comprises all other days. Estimates include country fixed effects; parentheses report t-statistics with standard errors clustered by calendar month. Benchmark availability yields 159,159 country-days for the individual legs and 162,587 for WML, so the reported coefficients need not add exactly. Construction details and individual country estimates are in Section IA.6 and Tables IA.2–IA.3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Momentum Returns by Holding Period K on PreTOM Window Days and Rest Days

	Loser leg		WML	
K	PreTOM days	Rest days	PreTOM days	Rest days
1	-47.9^{***}	-13.0	$+60.9^{***}$	$+35.8$
	(-3.7)	(-0.6)	(3.6)	(1.3)
3	-47.3^{***}	-10.9	$+57.5^{***}$	$+27.8$
	(-4.0)	(-0.5)	(3.8)	(1.0)
6	-43.2^{***}	-2.6	$+55.0^{***}$	$+17.5$
	(-3.8)	(-0.1)	(3.9)	(0.7)

Notes: The table reports returns for the $12/1/K$ WML momentum strategy of Jegadeesh and Titman (1993) over 1980–2025, where K stands for the holding period. Reported returns are equal-weighted averages of the K most recent momentum-portfolio cohorts, in basis points per month. Each cohort is a value-weighted momentum-decile portfolio formed at month-end and held for K months. The loser-leg return is market-adjusted. A day is classified as a PreTOM day if its calendar position lies in $[\tau-9, \tau-4]$; all other days are classified as Rest days. Because consecutive months share $K-1$ cohorts, the reported t-statistics use Newey–West standard errors with $K-1$ lags. The sample consists of 552 monthly observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Momentum Returns Before and After 2002

	(1) 1980–2001	(2) 2002–2025	(3) Later − earlier	(4) t
WML, PreTOM	10.24***	10.06**	−0.18	−0.03
	(3.11)	(4.59)		
WML, Rest days	7.01***	−1.95	−8.96**	−2.38
	(2.39)	(2.91)		
Losers − Mkt, PreTOM	−9.65***	−6.46*	+3.19	0.75
	(2.33)	(3.53)		
Losers − Mkt, Rest days	−4.14**	2.28	+6.41**	2.18
	(1.78)	(2.34)		
Change in WML, PreTOM − Rest days			+8.78	1.31
Change in Losers − Mkt, PreTOM − Rest days			−3.22	−0.63

Notes: Mean daily value-weighted returns are reported in basis points for two subsamples split at the end of 2001. The two subsamples contain 264 and 288 months, respectively. Loser-leg returns are market-adjusted, and WML denotes winners minus losers. PreTOM comprises days $[\tau-9, \tau-4]$, where τ denotes month-end; Rest days are all remaining days. For each month, daily returns are first averaged separately within the PreTOM and Rest windows. All reported means, standard errors, and t-statistics are then computed across these monthly averages. Standard errors are reported in parentheses below each period mean. Column (3) reports the change in the mean return from the earlier to the later subsample, and column (4) reports the corresponding t-statistic. For the final two rows, the change in Rest-day returns is subtracted from the change in PreTOM returns reported in column (3). These rows therefore measure the additional change in PreTOM returns relative to the change in Rest-day returns. Internet Appendix Table IA.4 provides a finer decomposition of the sample into five historical subperiods. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: The PreTOM Window's Abnormal Contribution to Momentum

Benchmark	Abnormal WML		Loser abnormal contribution	
	bps/ month (1)	Share of WML (2)	bps/ month (3)	Share of abnormal WML (4)
Panel A: Portfolio returns, temporal benchmark				
All other days (Rest)	+46.6** (2.2)	48%	+42.7*** (2.7)	92%
Mid-month days (Mid)	+45.3** (2.1)	47%	+30.3* (1.9)	67%
Panel B: Excluding the 2001–02, 2009 and 2020 crash episodes				
All other days (Rest)	+50.5*** (2.6)	52%	+45.9*** (3.1)	91%
Mid-month days (Mid)	+46.1** (2.2)	48%	+30.6** (2.0)	66%

Notes: Column (1) reports the average abnormal WML return during the PreTOM window, in basis points per month. The benchmark is the average daily WML return outside the PreTOM window, either on Rest or Mid days. Column (2) expresses the abnormal PreTOM return as a percentage of the average monthly WML return over the full sample. Column (3) measures the contribution of the loser leg to the abnormal PreTOM WML return. Specifically, it captures the additional underperformance of loser stocks relative to the market during PreTOM compared with the benchmark days. Column (4) expresses this contribution as a share of the abnormal WML return reported in column (1). The sample covers 1980–2025 and uses value-weighted momentum portfolios with fixed monthly sorting. Rest comprises all days outside the PreTOM window. Mid comprises the middle of the same calendar month, from the fourth trading day through the day before the PreTOM window begins. Panel B repeats the table excluding the momentum-crash episodes August 2001 to March 2002, January to September 2009, and February to May 2020. t-statistics in parentheses are from paired monthly contrasts: the window-minus-benchmark difference is computed month by month and its mean is tested against zero across months. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Bootstrap intervals, alternative benchmarks, and subperiod analyses are reported in Internet Appendix Table IA.26.

Table 7: The Loser Decile's Cumulative Path Around the Turn of the Month

	D1 (losers)	D2–D9	D10 (winners)
Window closes, $\tau-4$	-47.9^{***}	-0.4	$+13.0^{*}$
	(-3.71)	(-0.16)	(1.66)
Rebound peak, $\tau+4$	-33.7	$+2.4$	$+28.8^{**}$
	(-1.62)	(0.57)	(2.44)
$\tau+9$	-49.6^{**}	$+0.6$	$+36.3^{***}$
	(-2.00)	(0.13)	(2.69)

Notes: The table reports the cumulative market-adjusted returns of momentum decile portfolios, in basis points. Returns are cumulated from the first day of the PreTOM window through the days after month-end and then averaged across the 552 months from 1980–2025. The days after month-end follow the same stocks, with decile membership held at the month's sort. Returns are value-weighted using lagged market capitalization and market-adjusted as $r_i - r^m$. The PreTOM window ends at $\tau-4$, where τ denotes month-end. The rebound peak is defined as the trading day after the PreTOM window on which the cumulative loser return reaches its least-negative value. The cumulative paths end at $\tau+9$, the last day available for every month. Heteroskedasticity-robust t-statistics, computed from the monthly cumulative returns, are reported beneath each estimate. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Settlement Change Difference-in-Differences

Panel A: Cell Means (loser-decile market-adjusted return, bps per day)		
	Pre T+1 settlement	Post T+1 settlement
Trading days 5–8 of the month (control)	−3.51	7.04
$\tau-3$	−3.69	−31.36
$\tau-4$	−6.90	51.36
Panel B: Portfolio-Level DiD (early-month control)		
	(1) Pre-period: T+2 era only (Sep 2017–Apr 2024)	(2) Pre-period: full history (1980–Apr 2024)
$\beta_{\tau-4}$: $\mathbf{1}_{[\tau-4]}$	−22.25	−3.39
	(16.82)	(4.82)
$\beta_{\tau-3}$: $\mathbf{1}_{[\tau-3]}$	−19.30	−0.18
	(14.36)	(5.49)
π: T1Reform_t	7.78	10.55
	(21.31)	(18.67)
$\theta_{\tau-4}$: $\mathbf{1}_{[\tau-4]} \times \text{T1Reform}_t$	66.56*	47.71
	(38.53)	(34.53)
$\theta_{\tau-3}$: $\mathbf{1}_{[\tau-3]} \times \text{T1Reform}_t$	−19.10	−38.22
	(31.83)	(28.66)
$\theta_{\tau-4} - \theta_{\tau-3}$ (one-day shift)	85.66**	85.93**
	(39.62)	(33.64)
Observations	297	1,653

Notes: Estimates of equation (4), using value-weighted market-adjusted loser returns in basis points. Each month contributes observations for $\tau-4$, $\tau-3$ and the mean of trading days 5–8 of the same month. The post-period begins in June 2024; May 2024 is excluded. Pre-periods are indicated above the columns. Panel A reports cell means and Panel B regression coefficients. Parentheses report standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Settlement Change Difference-in-Differences (continued)

Panel C: Stock-level triple difference, D1 versus D10	(1) Pre-period: 1980–Apr 2024
$\theta_{\tau-4}$: $\text{Loser}_{i,t} \times \mathbf{1}_{[\tau-4]} \times \text{T1Reform}_t$	161.24**
	(73.23)
$\theta_{\tau-3}$: $\text{Loser}_{i,t} \times \mathbf{1}_{[\tau-3]} \times \text{T1Reform}_t$	−35.07
	(52.70)
$\theta_{\tau-4} - \theta_{\tau-3}$ (one-day shift)	196.32**
	(89.32)
Observations	4,907,237
Panel D: European T+3-to-T+2 transition, October 6, 2014	(1)
$\theta_{\tau-5} - \theta_{\tau-4}$: pooled, 13 markets (daily panel, month-clustered)	17.65**
	(8.82)
Inverse-variance weighted estimate†	14.73
Countries with the predicted sign	10 of 13
Placebo pair $\tau-7$ versus $\tau-6$	−1.15
	$p = 0.842$

Notes: Panel C estimates equation (5) on D1 and D10 stock-days, weighted by lagged capitalization, with stock and date fixed effects. Returns are market-adjusted, in basis points; standard errors are two-way clustered by stock and date. May 2024 is excluded and the post-period begins in June. Panel D reports the $\tau-5$ versus $\tau-4$ contrast of equation (6) for the thirteen European transition markets. The first row is the primary estimate: the pooled daily panel with country and calendar-month fixed effects and common slopes, with standard errors clustered by calendar month. †This row is an inverse-variance-weighted average of the separate country contrasts. It is reported as a descriptive magnitude only, with no standard error or test attached; Section 4.2 explains why. Country estimates are in Internet Appendix Section IA.17. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Net Selling Pressure on Losers, 2003–2022

	(1)	(2)
Loser (β_1)	1.22***	3.31***
	(20.20)	(38.90)
Loser × PreTOM (β_2)	0.20**	0.19**
	(2.50)	(2.22)
Stock fixed effects	Yes	No
Date fixed effects	Yes	Yes
Observations	17,861,276	17,861,277

Notes: Estimates of equation (7) on the CRSP/TAQ panel, 2003–2022. Net selling pressure is (sell volume − buy volume)/total volume in percent, with Lee–Ready trade classification from WRDS Intraday Indicators. PreTOM is $[\tau-9, \tau-4]$ and the omitted days are every day outside it. Parentheses report t-statistics with standard errors two-way clustered by stock and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Mutual-Fund Liquidation by Momentum Group, 1991–2024

Panel A: Raw rates by momentum group

	D1 (losers)	D2–D3	D4–D7	D8–D9	D10 (winners)
Share of positions closed (%)	14.3	10.2	8.9	9.6	12.0
Mean liquidation beyond pro rata (%)	5.4	2.7	2.4	3.2	4.5

Panel B: Within-fund-report differences

	Relative to D2–D9	
	D1 (losers)	D10 (winners)
Probability the position closes (Closure)	6.51^{***}	-0.18
	(16.66)	(-1.31)
Liquidation beyond pro rata (ExcessLiq)	4.53^{***}	-0.91^{***}
(continuous measure of liquidation)	(9.94)	(-4.57)

Notes: Panel A reports raw rates by momentum group. The closure rate is the share of positions absent from the fund's next holdings report, and positions in stocks delisted during the interval are excluded throughout. Panel B estimates equation (9) for winners and losers, with fund × report fixed effects and the middle deciles (D2–D9) as the omitted reference, so each coefficient is a difference from those deciles, in percentage points; t-statistics in parentheses, standard errors three-way clustered by fund, by stock and by report date. Liquidation beyond pro rata is the fraction of the initial position sold in excess of proportional scaling to the fund's flow, and a full exit is the limiting case. Under proportional selling, the departure from the flow benchmark caused by returns during the interval is common across a fund's holdings, so the Panel A levels contain a component common to all groups, which the fixed effects in Panel B remove. The sample is fund–stock–report observations from Thomson-Reuters mutual-fund equity holdings data, 1991–2024. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: Fund Outflows and the Selective Closure of Loser Positions, 1991–2024

	Baseline (1)	Position controls (2)	Within weight quintiles (3)
Loser × Outflow	0.0728***	0.0330***	0.0546***
	(4.66)	(2.77)	(5.66)
Winner × Outflow	−0.0181**	−0.0222***	−0.0103*
	(−2.49)	(−2.79)	(−1.91)
Loser × Inflow	0.0042	0.0033	0.0058*
	(0.47)	(0.36)	(1.70)
Winner × Inflow	−0.0112***	−0.0086**	−0.0032
	(−3.04)	(−2.54)	(−1.45)
Loser: Outflow − Inflow	0.0686***	—	—
	(4.19)		
Position controls	No	Yes	—
Estimated separately by weight quintile	No	No	Yes
Fund × report fixed effects	Yes	Yes	Yes
Observations	49,378,878	48,695,765	49,377,766

Notes: Estimates of equation (10) on fund–stock–report observations from mutual-fund equity holdings, 1991–2024. The dependent variable is an indicator that the position is absent from the next report. *Outflow* and *Inflow* are the magnitudes of net outflows and net inflows over the reporting interval, respectively. Each is zero when the net flow runs in the opposite direction. Position controls are the position's share of the fund's portfolio, the bid-ask spread, log size and log dollar volume, each as a main effect and again interacted with the outflow. Column (2) also includes the fund's trailing twelve-month return, measured to the report date that opens the interval and interacted with the loser and winner indicators. Column (3) estimates the specification separately within quintiles of initial portfolio weight; reported coefficients average the five quintile-specific estimates for the main interaction of interest, Loser × Outflow, which are individually positive and significant in every quintile, from 0.108 in the smallest positions to 0.035 in the largest. *Loser: Outflow − Inflow* tests whether the loser slope differs between money leaving and money arriving. Positions in stocks delisted during the interval are excluded. All statistics are equal-weighted across positions. t-statistics in parentheses, standard errors three-way clustered by fund, by stock and by report date. The share of fund-reports in net outflow is 60.0%. Internet Appendix Table IA.13 provides additional data. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: Distance from the 52-Week High within Momentum Losers

Quintiles by distance to 52-week high	Q1 nearest	Q2	Q3	Q4	Q5 farthest	Q5 − Q1
Panel A: Returns within momentum losers, 1980–2025						
Mean price / 52-week high	0.66	0.53	0.44	0.34	0.20	
PreTOM return, bps per day	−8.80	−11.48	−12.80	−12.81	−18.40	−9.60*** (−2.79)
Rest return, bps per day	+4.97	+5.52	+5.56	+4.38	+2.78	−2.19 (−0.82)
Panel B: Net selling pressure within momentum losers, percent of volume, 2003–2022						
Net selling, PreTOM	3.72	4.37	5.22	6.11	7.71	3.99*** (21.46)
Net selling, Rest	3.37	4.07	4.97	5.92	7.31	3.95*** (30.88)
Net selling, PreTOM − Rest	0.37** (2.00)	0.33* (1.89)	0.26 (1.45)	0.21 (1.13)	0.41** (2.38)	

Notes: Within the bottom momentum decile, stocks are sorted at the prior month-end by split-adjusted price divided by its trailing 252-day maximum (Jensen et al., 2023). Q1 is nearest the high, Q5 farthest. Panel A reports quintile means of value-weighted returns above the risk-free rate, in basis points per day, during PreTOM and Rest. Panel B reports net selling pressure, (sell volume − buy volume)/total volume in percent, from the WRDS Intraday Indicators, on 3,004,789 loser stock-days with both a classified imbalance and a predetermined 52-week-high ratio. PreTOM is $[\tau-9, \tau-4]$ and Rest comprises the other days. t-statistics for the Q5 − Q1 spreads are Newey–West with five lags on the daily spread. The PreTOM − Rest row differences the two rows above it within each calendar month and reports the mean of those monthly differences, with Newey–West t-statistics at three lags beneath. The two panels cover different periods because trade classification begins in 2003. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A Data Description

Table A.1 collects the coverage of every dataset the paper draws on and the moments of the stock panel by momentum group. Panel A gives the period each source covers, which is the sample of every estimate drawn from it. Panel B describes the three groups across which almost every contrast in the paper is drawn, including the two characteristics that carry the within-leg results, the quoted spread and distance from the trailing high.

Table A.1: Data Sources and Summary Statistics

Panel A. Datasets

Source	Period	Unit	Observations	Stocks	Funds
CRSP daily stocks, fixed monthly sorts	1980–2025	stock-day	53,324,149	22,744	–
the same build, historical	1927–2025	stock-day	73,648,246	25,597	–
WRDS Intraday Indicators (TAQ)	2003–2022	stock-day	17,861,277	9,416	–
Thomson S12 holdings, CRSP fund flows	1991–2024	fund-stock-report	49,378,878	15,397	11,185
Morningstar daily fund flows	2008–2023	fund-day	7,269,917	–	3,749
Compustat Global, ex-Japan markets	1991–2026	country-day	162,587	348	–
JKP, price over the 252-day high	1980–2025	stock-month	2,320,058	20,893	–

Panel B. Stock-level moments by momentum group

	D1 (losers)	D2–D9 (middle)	D10 (winners)	All deciles
Stocks per decile-month	881	390	614	4,615
Market capitalization, mean ($m)	512	4,092	2,870	3,249
Market capitalization, median ($m)	32	215	174	142
Share price, median ($)	3.07	16.74	16.12	13.11
Bid-ask spread, mean (%)	8.37	3.28	2.98	4.25
value-weighted mean	0.44	0.17	0.19	0.18
Dollar volume, median ($m per day)	0.081	0.493	0.737	0.355
Share turnover, mean (% per day)	1.677	0.542	0.961	0.813
value-weighted mean	1.345	0.649	1.193	0.733
Daily excess return, s.d. (%)	7.89	3.78	4.28	4.89
Net selling pressure, mean (% of volume), 2003–2022	5.27	1.97	1.39	2.45
value-weighted mean	−0.35	0.00	0.13	0.00
Price over 252-day high, mean	0.435	0.798	0.841	0.739
value-weighted mean	0.599	0.882	0.911	0.877
Price over 252-day high, median	0.430	0.840	0.889	0.801

Notes: Panel A lists the datasets the paper uses and the coverage of each. The CRSP panel is the fixed-monthly-sort build as described above, and the 1927–2025 row is that same build before the 1980 start of the baseline sample. This longer sample supports only the historical evidence of Section IA.7. The 1980–2025 count is the baseline regression sample of Table 2. Sell pressure is merged in from the WRDS Intraday Indicators, and its count is the number of panel rows carrying a non-missing value. Fund positions are Thomson S12 holdings paired on consecutive report dates and matched through MFLINKS to CRSP implied flows, on the estimation sample of Table 10, which holds 371,689 fund-reports. Daily fund flows are from Morningstar for U.S. equity open-end funds, excluding funds with an exchange-traded share class. The international panel is aggregated to country-days, so its stock entry is the average number of stocks behind a country-day rather than a count of securities, and the monthly JKP characteristic is counted in the panel stock-months it matches. Panel B covers 1980–2025, except net selling pressure, which covers 2003–2022. Panel B is equally weighted across stock-days within each group, except the value-weighted rows, which weight each stock-day by its lagged market capitalization as the portfolios do, and each row label names the statistic reported, since capitalization, price and volume are heavily right skewed. Market capitalization is the lagged value that supplies the portfolio weights. The spread is the quoted spread over the quote midpoint, where the quote is not crossed, and turnover is dollar volume divided by price and by shares outstanding. Only return dispersion is given, since an equally weighted mean across stock-days is dominated by the smallest and most volatile stocks. Net selling pressure is sell volume minus buy volume over total volume, so positive is net selling. Price over the trailing high is JKP's `prc_highprc_252d` at the end of the preceding month, as in Table 12.

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